2024

SOFTWARE REQUIREMENT SPECIFICATION (SRS)

FINAL YEAR PROJECT (FYP) SUPERVISOR HUNTING APPLICATION

LIST OF AUTHORS' NAMES
MIRZA RUSYAIDI BIN RAHAMAT CB23124
NABILAH BINTI MOHAMMAD SHAHIDAN CB23122
NUR ASYIKIN BINTI ISMAIL CB23121
AMALA KARTHIGAYAN CB23116
Supervisor Hunting System
Submitted to the Software Requirement Workshop
Bachelor of Computer Science (Software Engineering) with Honours
Faculty of Computing,
Universiti Malaysia Pahang Al-Sultan Abdullah

DOCUMENT APPROVAL

	Name	Date
Authenticated by: mirzarusyaidi Project Leader	MIRZA RUSYAIDI BIN RAHAMAT	14 May 2024
Approved by: nabilah Project Manager	NABILAH BINTI MOHAMMAD SHAHIDAN	14 May 2024
Approved by: ________________ Client	PM TS. DR. ROHANI BINTI ABU BAKAR	14 May 2024

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1. INTRODUCTION

1.1 PURPOSE

The purpose of this document is to outline the specifications and requirements for the development of the Supervisor Hunting Application, designed to assist in the management of final year project (FYP) supervision for undergraduate students in the Faculty of Computing. The purpose of this document is to provide a comprehensive Software Requirements Specification (SRS) for the Final Year Project (FYP) Supervisor Hunting Application, an application proposed by the Faculty of Computing, University Malaysia Pahang Al-Sultan Abdullah. The aim of this document is to provide an overview of all parties associated with the project that all expected readers, which are developers, owner of the system (Faculty of Computing), respective stakeholders such as students, lecturers, Final Year Project coordinators, head of research group, and team members have to make sure that everyone is aware of their obligations and functions, it sets forth the objectives and requirements of the system.

The document encompasses the Software Requirements Specification (SRS) and the Request for Proposal (RFP). The SRS section ensures that the development team and stakeholders have a clear and common knowledge of what the system is designed to achieve by providing a full explanation of the functional and non-functional requirements of the system. In the RFP section, offers that can fulfil the requirements and goals of the project are welcomed, along with the criteria for choosing outside suppliers or partners, if necessary. This document's scope includes outlining project goals, like making it simpler for students can manage the supervisor hunting process and make student registration, the lecturer can post and manage interest FYP topics, manage appointment for student, the coordinators can manage the hunting schedule, announcement, manage quotas and generate supervisor hunting reports for coordinators. Then, other scopes are the system can manage notifications, giving real-time information on supervisor availability and Dean can view and monitor activities within the system.

1.2 SYSTEM IDENTIFICATION

The Software Requirement Specification (SRS) belongs to the "FYP Supervisor Hunting Application".

Table 1.1 System Identification

<table>
<tr><td>System Identification Number</td><td>1000 (SRW_SRS_FYPSHA 1.1.24)
The system identification number shows the identification number for Final Year Project (FYP) Supervisor Hunting Application</td></tr>
<tr><td>System Title</td><td>Final Year Project (FYP) Supervisor Hunting Application</td></tr>
<tr><td>System Abbreviation</td><td>FYPSHA</td></tr>
<tr><td>System Version Number</td><td>1.1.24
This is initial release number for Final Year Project (FYP) Supervisor Hunting Application</td></tr>
<tr><td>System Release Number</td><td>1.1.24
Start with number 1as the major release number and middle with number 1 as the minor release number that this is initial version of the system.
Number 24 as the service release that shows the system released in 2024.</td></tr>
<tr><td>Document Identification ID</td><td>SRS_FYPSHA_2024
Meaning of term:<table><tr><td>SRS</td><td>SOFTWARE REQUIREMENT SPECIFICATION</td></tr><tr><td>FYPSHA</td><td>Final Year Project Supervisor Hunting Application (name of system)</td></tr><tr><td>2024</td><td>Year of document created</td></tr></table></td></tr>
<tr><td>Use Case ID</td><td>MCAUC100_FYPSHA_2024
Meaning of term:<table><tr><td>MCAUC</td><td>Manage Coordinator Activities Use Case</td></tr><tr><td>100</td><td>Number of use case in the system</td></tr><tr><td>FYPSHA</td><td>Final Year Project Supervisor Hunting Application (name of system)</td></tr><tr><td>2024</td><td>Year of document created</td></tr></table></td></tr>
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1.3 SYSTEM OVERVIEW

The proposed Supervisor Hunting Application is designed to streamline the process of students in the Faculty of Computing finding supervisors for their Final Year Projects (FYP). This system aims to assist the FYP Coordinator in managing interactions between students and lecturers, ensuring an organized and efficient matching process based on project themes, student interests, and lecturer expertise.

The application serves various users: the Deputy Dean of Research, FYP Coordinator, Head of Research Group (HoRG), FC Lecturers, and FC Students. Key features include setting schedules and announcements for supervisor hunting activities, managing supervisor quotas, allowing lecturers to post FYP topics, enabling students to search and communicate with potential supervisors, and registering agreed supervisors. The system also provides real-time updates on supervisor availability, suggests relevant supervisors for students who haven't identified one, and allows the HoRG and Coordinator to monitor the process and generate reports.

Overall, the Supervisor Hunting App enhances efficiency, transparency, and communication in the supervisor selection process, ensuring every student finds a suitable supervisor for their FYP.

1.4 REFERENCES

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2. PRODUCT DESCRIPTION

2.1 Product Perspective

Illustrates the context diagram of the FYP Supervisor Hunting Application. A detailed description of the data flows is explained below.

1. Coordinator able to set the schedule and do the announcement regarding the supervisor hunting activities.

   Data Flow:

   - Supervisor Selection Schedule: The coordinator sends the schedule data to the system.
   - Hunting Announcement: The coordinator sends announcements to the system.

   Interaction: The coordinator inputs the schedule and announcements into the system, which disseminates this information.

2. Coordinator able to set and update the quota of supervisee for each lecturer.

   Data Flow:

   - Lecturer Quota: The coordinator updates lecturer quotas in the system.

   Interaction: The coordinator manages the quota for each lecturer through the system.

3. Lecturers able to post their interest topic for Final Year Project (FYP).

   Data Flow:

   - Final Year Project (FYP) Topics Data: Lecturers input their FYP topics into the system.

   Interaction: Lecturers use the system to enter their areas of interest for FYP.

4. Students are able to search and communicate with their prospective supervisor regarding their interest in Final Year Project (FYP)..

   Data Flow:

   - Supervisor Search: Students perform searches for prospective supervisors in the system.
   - Student Message and Lecturer Message: Students and lecturers communicate through the system.

   Interaction: Students use the system to search for supervisors and send messages to them. Lecturers respond via the system.

5. Students are able to register their agreed supervisor.

Data Flow:

- Agreed Supervisor Registration: Students register their chosen supervisors in the system.

Interaction: Once a student and supervisor agree, the student registers the selection in the system.

6. System should be able to update in real-time the available quota for each lecturer.

   Data Flow:

   - Implicit in the system's operations; real-time updates ensure accurate quota management based on student registrations.

   Interaction: The system automatically updates lecturer quotas as students register their supervisors.

7. The system should be able to suggest any relevant supervisor or relevant research group for the student if the student is not able to indicate any prospective supervisor.

   Data Flow:

   - Supervisor Suggestions: The system suggests relevant supervisors to students.

   Interaction: The system provides suggestions to students who do not have a prospective supervisor based on their interests.

8. HORG and coordinator should be able to monitor supervisor hunting by generating relevant reports.

   Data Flow:

   - Supervisor Hunting Report: Generated by the system and accessible to the coordinator and HORG.
   - Hunting Activity Monitoring: Sent to Dean for monitoring purposes.

   Interaction: The system generates reports that HORG and coordinators can access to monitor the process.

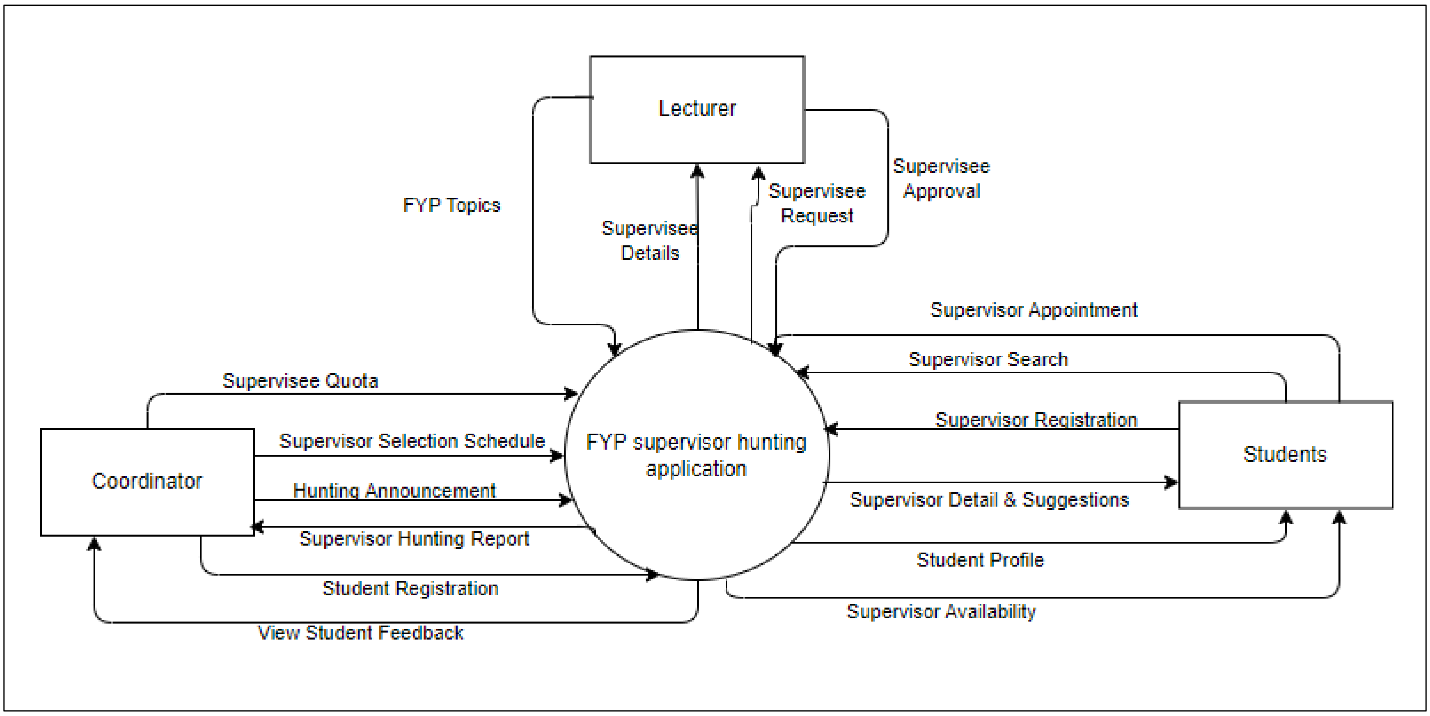


Figure 2.1 Context diagram of the Final Year Project (FYP) Supervisor Hunting Application

2.2 Product Functions

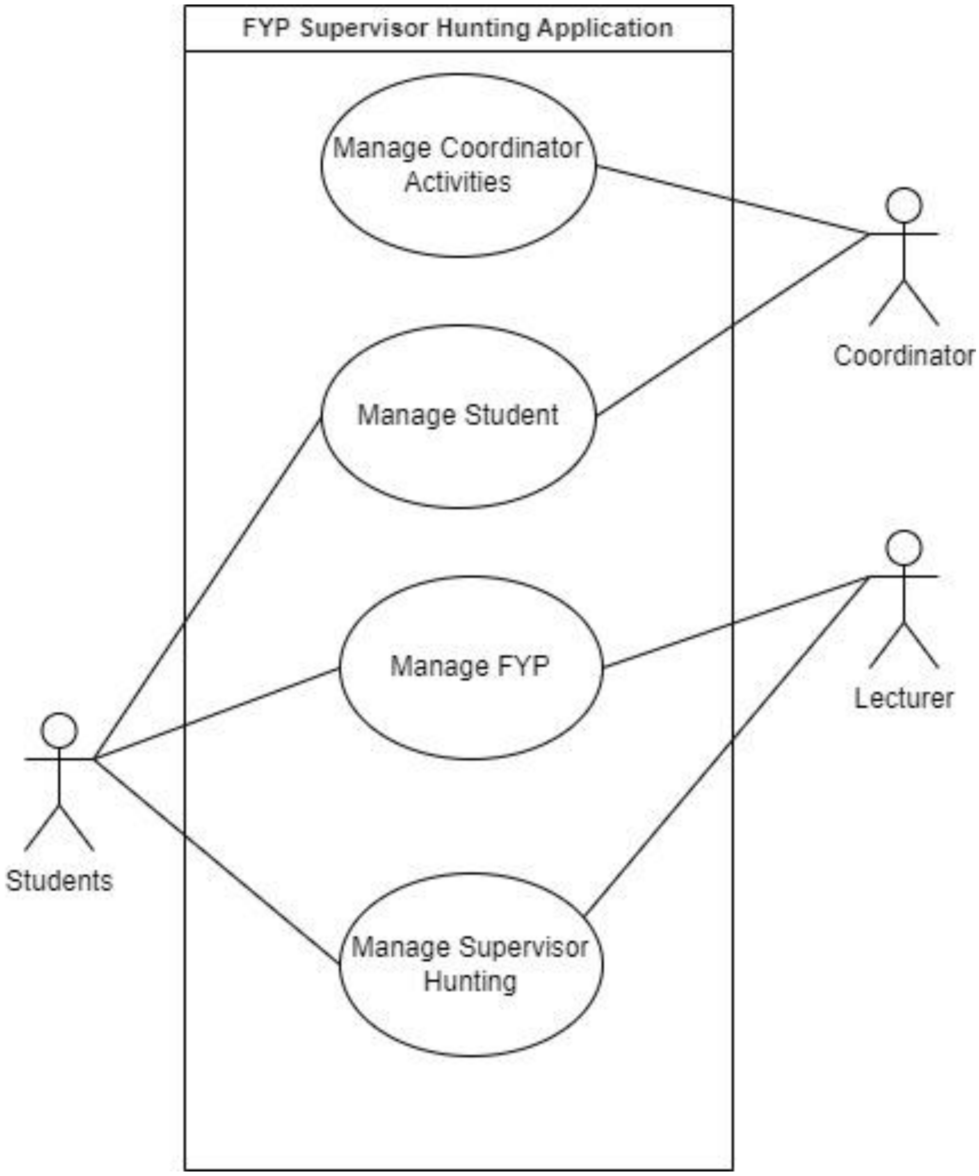


Figure 2.2: Use Case of the Final Year Project (FYP) Supervisor Hunting Application

2.3 User Characteristics

Table 2.1 User Characteristics of the stakeholders

User	Educational Level	Experience	Technical Expertise
Final Year Project (FYP) Coordinator	Master in Computer Science or a related field	Several years of experience in academic coordination, teaching, and student supervision.	Proficient in project management tools, student information systems, and coordination of academic programs.
Lecturer	Master in Computer Science or a related field	Several years of teaching and supervising undergraduate and/or postgraduate projects.	Proficient in the subject matter they teach, experience with academic supervision, and familiarity with learning management systems.
Student	Undergraduate students in their final year of study.	Basic understanding of research methodologies and technical skills acquired during their FYP's.	Competence in relevant computing skills, familiarity with project management basics, and ability to use academic and communication platforms.

3. SPECIFIC REQUIREMENTS

3.1 Software Product Features

3.1.1 Manage Coordinator Activities Use Case (NUR ASYIKIN)

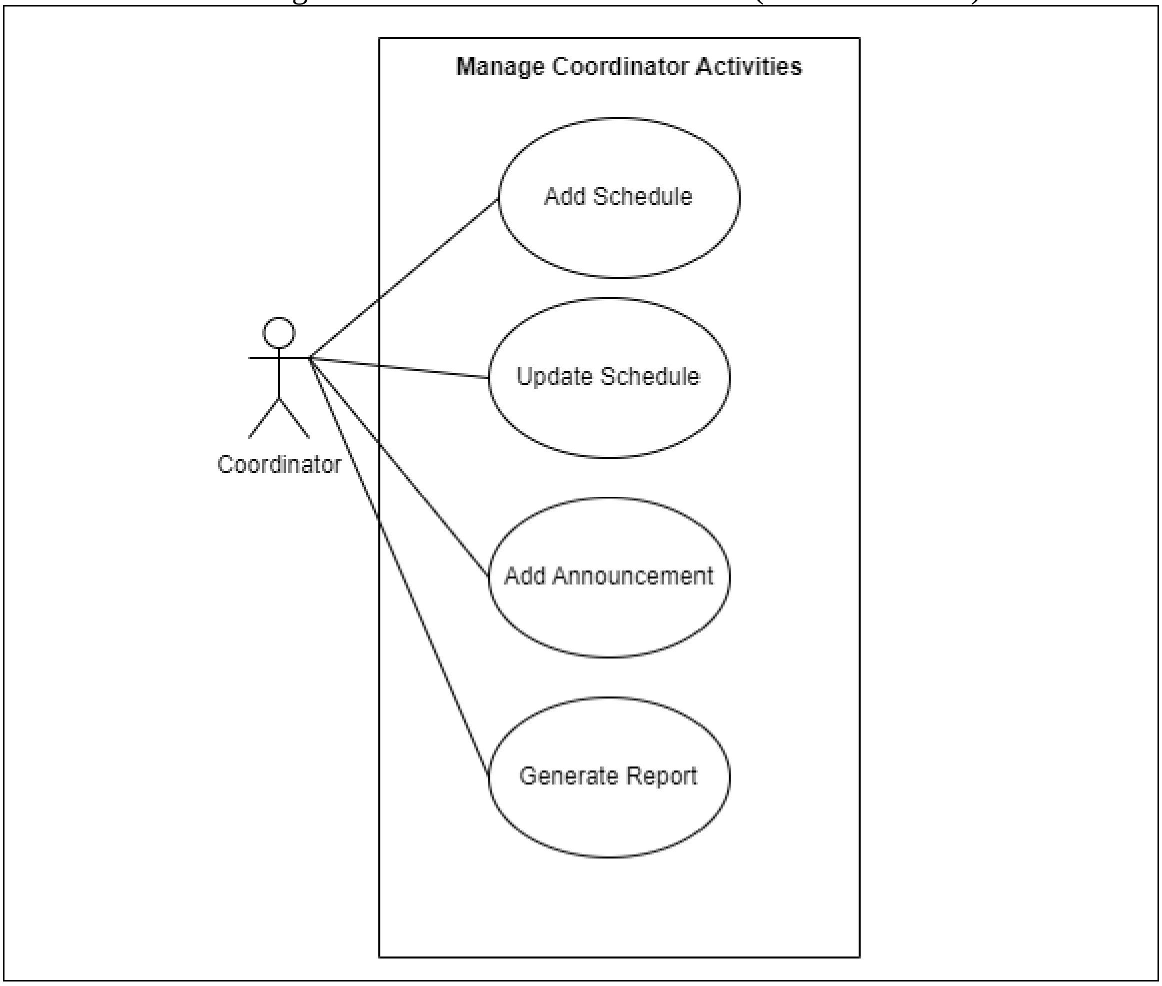


Figure 3.1 Manage Coordinator Activities Diagram

Table 3.1 Manage Coordinator Activities Use Case Description

Use Case ID	MCAUC100_FYPSHA_2024
Brief Description	This use case allows the coordinator to add schedule, update schedule, delete schedule, add announcement and generate report to accommodate the coordinators activities.
Actor	Coordinator
Pre-Conditions	1. The coordinator must be logged into the system.
Basic Flow	The Use case starts when the coordinator is assigned to the Manage Coordinator activities dashboard by the system.

	1. The coordinator will see different buttons to click which are [A1: Add Schedule] to go to schedule interface. [A3: Add announcement] to go to the announcement interface. [A4: generate the supervisor hunting report] to go to report interface. 2. The system saves changes that have been made. 3. The system generates a message box. 4. Display the message box to the coordinator. 5. The use case end.
Alternative Flow	Alternative Flow A1: Add Schedule (**ReqTrac101_FYPSHA_2024**) 1. The system displays the schedule timeline interface. The coordinator is shown [A1. Add Schedule] by entering dates and inserting the time details to set the schedule. The coordinator can update the schedule if necessary [A2. Update Schedule]. 2. The coordinator enters the schedule details in the form of date and time**.** 3. The coordinator clicked button <<Add schedule>> to add schedule**.** 4. The system database saves the schedule details. 5. The system prompt user that details are successfully update the message. 6. The use case continues at basic flow step number 2. A2: Update Schedule 1. The coordinator clicks on the <<Update schedule>> button. 2. The coordinator enters the new schedule details in the form of date and time**.** 3. The coordinator clicks on the <<Add schedule>> button. 4. The system saves the updated schedule details. 5. The use case continues to step number 2 in basic flow. A3: Add Announcements (**ReqTrac102_FYPSHA_2024**) 1. The coordinator clicks the button <<add announcements>> to add announcement. 2. The coordinator inputs the announcement details and duration of the announcement.

	3. The coordinator clicks button <<Publish>> to display the announcement. 4. The database saves the changes of the system. 5. The system prompts a successfully published message. 6. The use case continues at basic flow step number 2. A4: Generate the supervisor hunting report (**ReqTrac103_FYPSHA_2024**) 1. The coordinator clicks the button <<Generate Report>> to Generate the supervisor hunting report. 2. The system displays the report. 3. The coordinator clicks button <<Save>> to save report file in device. 4. The database saves the changes of the system. 5. The system prompts a successfully save message. 6. The use case continues at basic flow step number 2.
Exception Flow	Exception Flow E1: Report Generation Error 1. The system displays an error message indicating the failure. 2. The system logs the error details for troubleshooting. 3. The system prompts the user to try generating the report again through the message box. 4. The use case continues at basic flow step number 6.
Post-Conditions	1. The system notifies the coordinator that the schedule or announcements have been successfully updated via message box.
Rules	1. Only users with Coordinator or HoRG roles can access this module.
Constraints	1. The announcement needs to be not more than 500 words per announcement.
Sequence Diagram	Refer Appendix A-1.1: Sequence Diagram – Basic Flow A-1.2: Sequence Diagram – Alternative Flow A-1.3: Sequence Diagram – Exception Flow

3.1.2 Manage Student (AMALA KARTHIGAYAN)

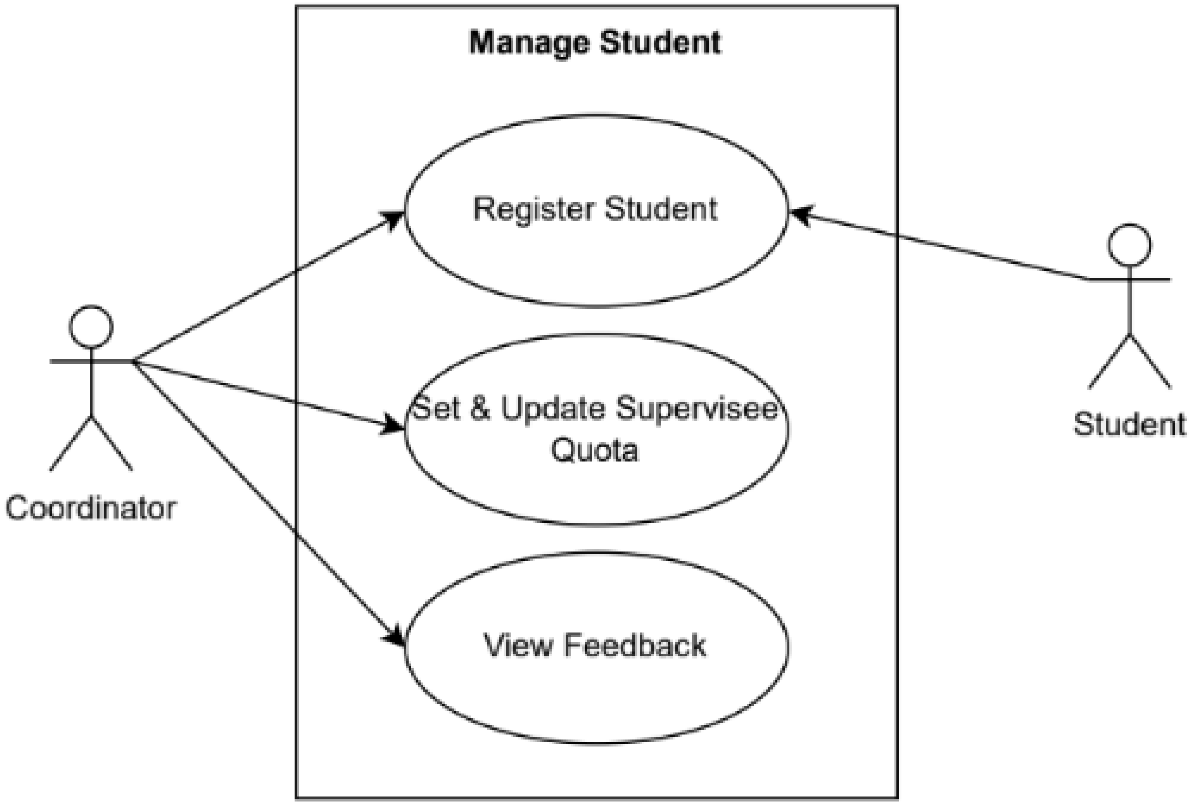


Figure 3.2: Manage Student

Table 3.2: Manage Student Use Case Description

Use Case ID	MSUC200_FYPSHA_2024
Brief Description	This use case allows the Coordinator to register students who have pre-registered for the Final Year Project (FYP) 1 subject, providing them with system access, set and update supervisee quota for each lecturer and view feedback given by students about the system.
Actor	Coordinator and Student
Pre-Conditions	1. The user must be logged into the system as a Coordinator.
Basic Flow	1. Use case begins when the Coordinator logs into the system and the system displays the Coordinator's Dashboard. 2. The coordinator has the option to: **[A1: Register Students]** ***(ReqTrac201_FYPSHA_2024)*** **[A2: Add Supervisee Quota]** ***(ReqTrac202_FYPSHA_2024)*** **[A3: View Feedback]** ***(ReqTrac205_FYPSHA_2024)*** 3. Coordinator is able to view the student registration list.

	4. Coordinator is able to view the current list of supervisors and their respective supervisee quota. 5. The system displays a list of feedbacks from the students. 6. The use case ends.
Alternative Flow	**A1: Register Students *(ReqTrac201_FYPSHA_2024)*** 1. The system displays the student registration page consists of the previous batches of registered students. 2. The coordinator has the option to: **[A.1.1: Create New Batch of Students]** **[A.1.2: Update Batch]** **[A.1.3: Delete Batch]** 3. Coordinator clicks on the <<Batch Icon>> that has just been created. 4. System displays the new batch page and shows a "No students registered" message. 5. Coordinator clicks the **[A.1.4: Import Document]** button to upload the list of pre-registered student details that has been accessed from the Open Registration system. 6. The system shows a list of students details imported from the document. 7. The Coordinator selects the number of students and clicks **[A.1.5: Generate Credentials]** button to create unique UserID and passwords for the students. 8. Use case return to step number 3 in the basic flow. **A.1.1: Create New Batch of Students** 1. System display a pop up page for the "New Batch Registration Form". 2. Coordinator fills in the details of the new batch **[E1: Invalid Batch Name]**. **3.** Use case returns to step number 3 in **A1: Register Students.** **A.1.2: Update Batch** 1. Coordinator clicks on the <<Update>> button. 2. Coordinator edit the batch details that she/he key-in before this

and click <<Confirm>> button.

3. Use case returns to step number 3 in **A1: Register Students.**

A.1.3: Delete Batch

1. Coordinator clicks on the <<Delete>> button.
2. The system displays a pop up confirmation message to delete the batch.
3. Coordinator clicks <<Delete>> button to confirm the deletion process.
4. Use case returns to step number 1 in **A1: Register Students.**

A.1.4: Import Document

1. System prompts the coordinator to upload the document containing list of pre-registered student details.
2. Coordinator upload the document **[E2: Invalid File Format].**
3. System validates the data in the document **[E3: Data not available].**
4. Upon successful validation and document import, the system displays "File uploaded successfully" message.
5. Use case return to step number 6 in **A1: Register Students.**

A.1.5: Generate Credentials

1. System generates a unique UserID and password for each pre-registered student in the list.
2. Upon successful credentials generation, the system displays "Credential generations successful" message.
3. Use case returns to step number 8 in **A1: Register Students.**

A2: Add Supervisee Quota *(ReqTrac202_FYPSHA_2024)*

1. Coordinator clicks on the <<Add & Edit Quota>> button for the selected supervisor.
2. Coordinator fills in the supervisee quota for the selected supervisor.
3. System update the quota details in the Quota database.
4. System displays a success message.
5. Use case returns to the step number 4 in the basic flow.

	A3: View Feedback *(ReqTrac205_FYPSHA_2024)* **1.** System retrieves and displays the feedback list **[E4: No Feedback Available].** 2. The coordinator reviews the feedback and mark the reviewed feedbacks. 3. Use case returns to step number 5 in the basic flow
Exception Flow	**E1: Invalid Batch Name** 1. The system validates the batch name entered by the Coordinator. 2. If the batch name is found to be invalid (e.g., it contains special characters, is too short or duplicates an existing batch name), the system displays an error message indicating the specific issue with the batch name. 3. The Coordinator corrects the batch name as per the guidelines provided by the system. 4. The system re-validates the corrected batch name. 5. Use case ends. **E2: Invalid File Format** 1. The system checks the format of the uploaded document. 2. If the file format is invalid (e.g., not a CSV or xlsx), the system displays an error message indicating the file format issue. 3. The Coordinator reviews the error message and corrects the file format as per the system requirements. 4. The Coordinator re-uploads the document. 5. The system re-validates the re-uploaded document. 6. Use case ends. **E3: Data not available** 1. The system validates the data within the uploaded document. 2. If any required data is missing or incomplete (e.g., student names, IDs, etc.), the system displays an error message specifying the missing or incomplete data. 3. The Coordinator reviews the error message and updates the

	document to include the missing or incomplete data. 4. The system re-validates the re-uploaded document. 5. Use case ends. **E4: No Feedback Available** 1. System displays a message indicating that "There's no feedback to display". 2. Coordinator goes back to dashboard. 3. Use case ends.
Post-Conditions	1. All pre-registered students for FYP 1 are successfully registered in the system with their unique credentials generated and sent via email. 2. The supervisee quotas for lecturer are updated in the system. 3. Feedback from students has been reviewed and marked as reviewed by the coordinator, with notifications sent to respective students.
Rules	1. The system must ensure that all data entered (e.g., batch names, supervisee quotas) meets validation criteria and is stored correctly without conflicts. 2. Only students who have pre-registered for the Final Year Project (FYP) 1 can be registered in the system.
Constraints	Quota adjustments must not exceed the set limits ***(ReqTrac203_FYPSHA_2024)*** and should reflect the actual availability of supervisors in real time ***(ReqTrac204_FYPSHA_2024).***
Sequence Diagram	Refer Appendix A-2.1: Sequence Diagram – Basic Flow A-2.2: Sequence Diagram – Alternative Flow A-2.3: Sequence Diagram – Exception Flow

3.1.3 Manage Final Year Project (MIRZA RUSYAIDI)

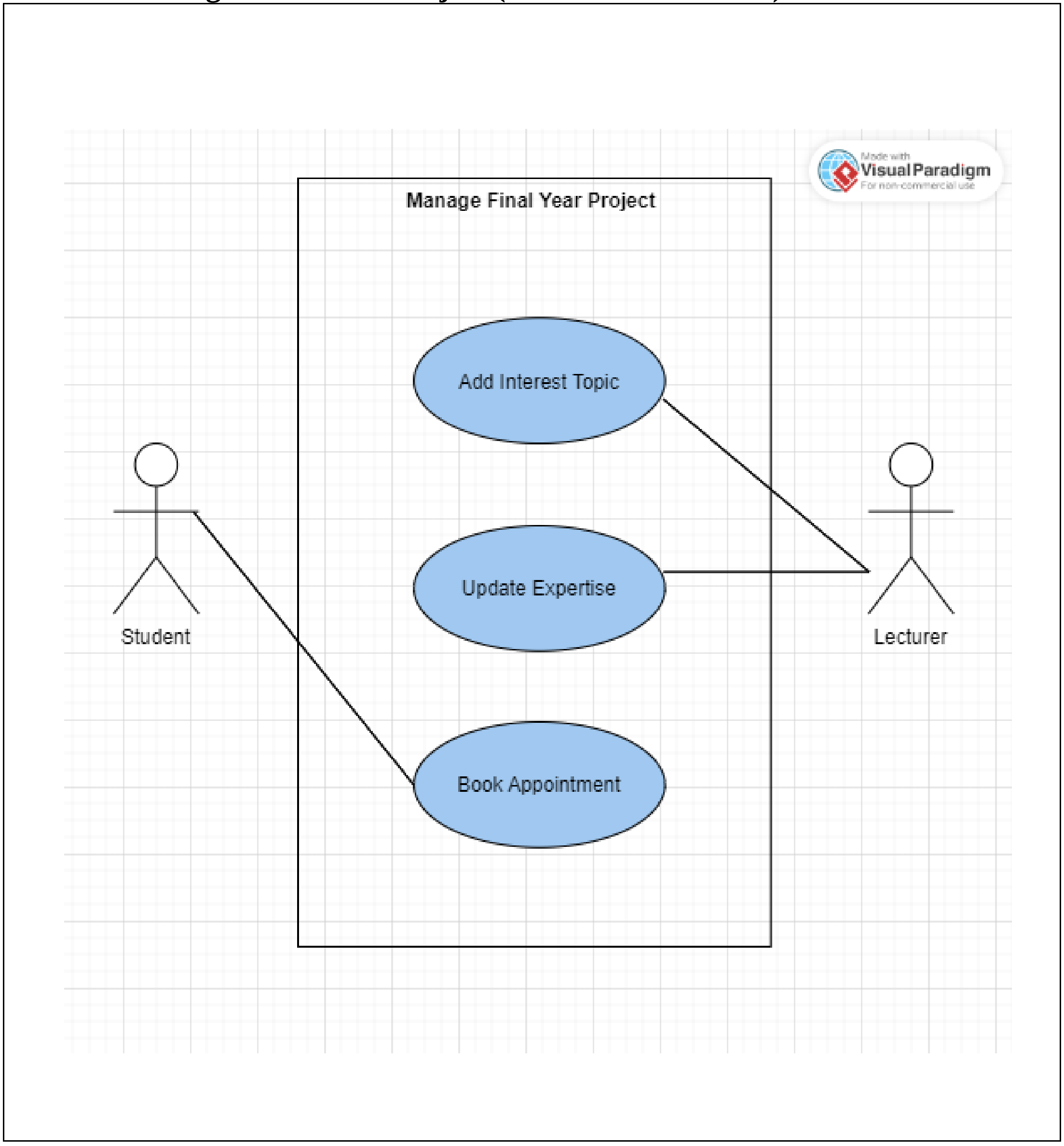


Figure 3.3 Manage Final Year Project Diagram

Table 3.3 Manage Final Year Project

Use Case ID	MFYPUC300_FYPSHA_2024
Brief Description	The Manage Final Year Project use case allows lecturers to add, update, view and delete their interest Final Year Project (FYP) topics and their expertise. Student can book appointment with the respective lecturer and lecturer can decide either to approve or reject the appointment.
Actor	Student, Lecturer
Pre-Conditions	1. The lecturer must be logged into the system as lecturer roles. 2. The student must be logged into the system as student roles.
Basic Flow	The use case begins with:

	Lecturer 1. The system display the lecturer profile. **[A1: Update Expertise]** The system displays the summary of the dashboard that contains registered supervisee, interest topics and supervisee request pending. **[A2: Show Registered Supervisee Details]**, **[A3: Show FYP Interest Topic], [A4: Show Supervisee Request Approval]** 2. The use case end. Student 1. The system shows Supervisor Details page. **[A5: Book Appointment]** 2. The use case end.
Alternative Flow	[A1: Update Expertise] 1. The lecturer can update their expertise at Lecturer Profile. 2. The lecturer clicks on <<Update>> button. 2. The use case continue at number 1 in basic flow. [A2: Show Registered Supervisee Details] 1. The lecturer can view the details of registered supervisee. 2. The use case continue at number 1 in basic flow. [A3: Show FYP Interest Topic] 1. The lecturer can add FYP interest topics by fill in the FYP Interest Topic Form. 2. The lecturer clicks on <<Submit>> button. 3. The lecturer can make a change of the topic, if necessary, by clicks on FYP Interest Topic Update form. 4. The lecturer can delete the topic. 5. The use case continue at number 1 in basic flow. [A4: Show Supervisee Request Approval] 1. The lecturer views pending supervisee request at Supervisee Request Pending page. 2. The lecturer can decide either to approve or reject the supervisee request. **[A6: Supervisor Approval]** 3. The use case continue at number 1 in basic flow.

	[A5: Book Appointment] 1. The student clicks the <<Book Appointment>> button. 2. The system shows the Book Appointment form. 3. The student can fill in the Book Appointment form. 4. The student click on submit button. **[E1: Cancel submit]** 5. The system will automatically send a notification via email to the respective lecturer about the appointment request. 6. The lecturer need to decide either to approve or reject the appointment. **[A7: Appointment Approval]** 7. The use case continue at number 1 in basic flow. [A6: Supervisor Approval] 1. The lecturer approved the request if the student agree on the interest topic provide by the lecturer. **[E2: Reject Request]** 2. The system automatically registered the student under selected supervisor. 3. The use case continue at number 1 in basic flow. [A7: Appointment Approval] 1. The lecturer approved the request of appointment within 3 days working hours. **[E3: Auto Cancelled]** 2. The system automatically send a notification via email to student regarding the appointment. 3. The use case continue at number 1 in basic flow.
Exception Flow	[E1: Cancel submit] 1. The student click on cancel button. 2. The use case end. [E2: Reject Request] 1. The lecturer reject the request. 2. The system send the notification via email regarding rejection status. 3. The use case return to step 1 in basic flow. [E3: Auto Cancelled] 1. The system auto cancelled the appointment if the lecturer did not respond within 3

	working days. 2. The use case end.
Post-Conditions	1. The lecturer has successfully update interest FYP topics and update their expertise into Final Year Project database. 2. The student has book appointment with a desired lecturer to discuss regarding final year project.
Rules	None
Constraints	1. Actions are restricted based on user roles (student or lecturer). 2. Only lecturer can add FYP interest topic.
Sequence Diagram	Refer Appendix A-2.1: Sequence Diagram – Basic Flow A-2.2: Sequence Diagram – Alternative Flow A-2.3: Sequence Diagram – Exception Flow

3.1.4 Manage Supervisor Hunting (NABILAH)

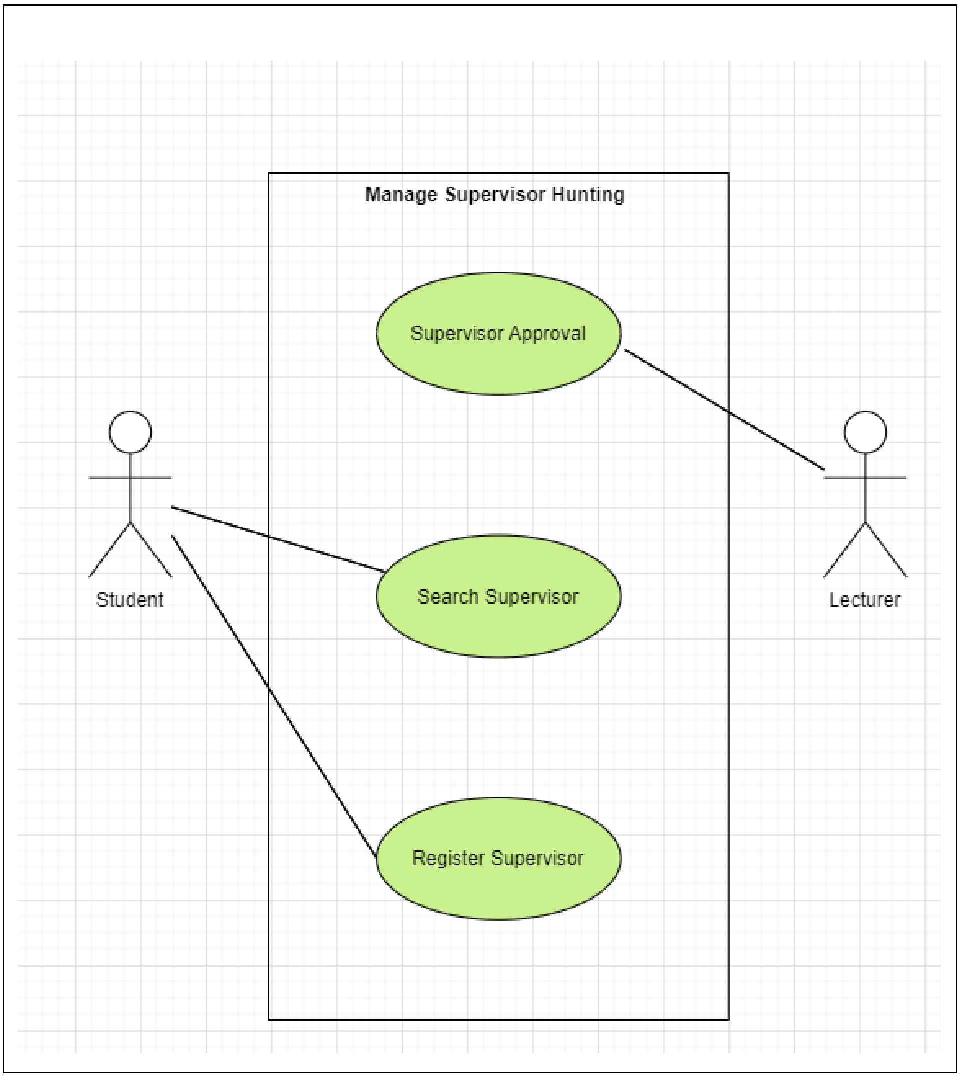


Figure 3.4 Manage Supervisor Hunting Diagram

Table 3.4 Manage Supervisor Hunting Use Case Description

Use Case ID	MSHUC400_FYPSHA_2024
Brief Description	This use case allows the student to search, sort the research group interests and register the supervisor for their Final Year Project (FYP). Students can search for prospective supervisors and sort based on research group regarding their interest's topic for their Final Year Project (FYP) and submit requests to register as a supervisee. Lecturers can approve or reject these requests.
Actor	Student, Lecturer

Pre-Conditions	1. The student able to access the information the list of lecturers that supervisors have been updated. 2. The lecturer able to approve or reject the supervisor registration requests from student.
Basic Flow	The use case begins with: Student 1. The system redirects to the “Student Dashboard” section. **2.** The student searches prospect supervisors by lecturer name. **[ReqTrac401_FYPSHA_2024]** 3. The student enters name of lecture regarding their interest topic for Final Year Project (FYP). **[A1: No Matching Supervisor Found]** 4. The student sorts based on research group regarding their interest topic for Final Year Project (FYP). **[ReqTrac402_FYPSHA_2024]** 5. The student selects any research group regarding their interest topic for Final Year Project (FYP). 6. The system displays a list of supervisors matching the student's interests. 7. The student selects a lecturer from the list. 8. The student fills in the information for the supervisor registration request form such as propose FYP title and brief description. **[A2: Supervisor Registration]** 9. The use case ends. Lecturer 1. The system redirects to the “Lecturer Dashboard” section. 2. The lecturer navigates to the “Supervisor Request Pending” page. **[A3: Approval Registration]** 3. The use case ends.
Alternative Flow	[**A1: No Matching Supervisors Found**] 1. The student may encounter no matching supervisors. 2. The system displays “No Record Found” message. 3. The process continues to step 2 from basic flow of Student Flow. **[A2: Supervisor Registration]** **1.** The system displays the request form for student who want to register the supervisor for their Final Year Project (FYP).

	2. The student fills in the information for the supervisor registration request form such as << **Title, Brief Description**>>. **3.** **[ReqTrac403_FYPSHA_2024]** **4.** The student clicks “Submit” button to register the chosen lecturer. **5.** The system notified the student's request via email. **6.** The process continues to step 1 from Basic Flow of Lecturer Flow. **[A3: Approval Registration]** **1.** The system displays the list of pending requests registration from student. **2.** The lecturer can decide either to approve or reject the supervisor registration requests from student. **3.** The lecturer clicks “Save” button to save the application. **4.** The system displays “Successfully Save” message. **5.** The system notifies the student via email about supervisor registration requests of approved. **6.** The system automatically registers the student as supervisee. **7.** The process continues to step 1 from Basic Flow of Student Flow.
Exception Flow	None
Post-Conditions	1. The supervisor's list of students is updated database to include the new student.
Rules	1. The student can only have one registered supervisor for their Final Year Project (FYP).
Constraints	1. Each lecturer can only supervise a limited number of students. 2. The lecturer can only approve requests up to their supervisory capacity such as the number of students they can supervise. 3. The function of search can be filter by lecturer name. 4. The supervisor application requests should be submitted and approved or rejected by chosen lecturer within a specific time frame such as before the start of the semester.
Sequence Diagram	A-4.1: Sequence Diagram – Basic Flow A-4.2: Sequence Diagram – Alternative Flow

3.2 External Interface Requirements

3.2.1 User Interfaces

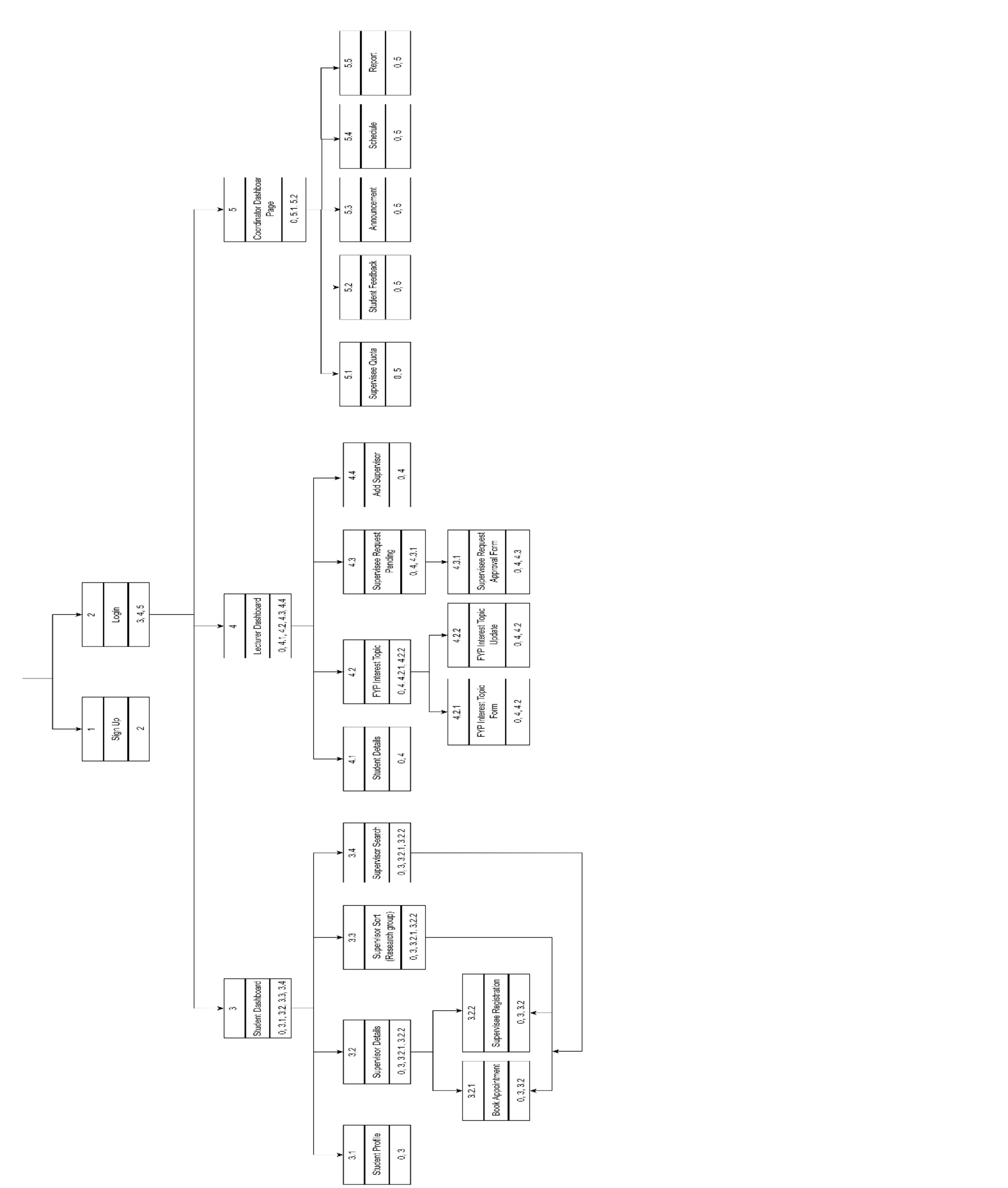


Figure 3.5 Dialogue Diagram

Table 3.5 User Interfaces Description

User Interface Description for Manage Coordinator Activities (NUR ASYIKIN)

User Interface Name or Number	Description	User Interface Layout
Coordinator activities	From this page coordinator can go to add announcement, schedule, generate report	Refer Appendix B.1
Add announcement	The coordinator can add announcement details here	
Schedule	The coordinator can add schedule and delete schedule in this interface	
Generate Reports	The coordinator can generate report and save report file in this interface	
Save Report	Save Report Notification	
Report Generation Error	The Coordinator Will be Notify on Report Generation Error	

User Interface Description for Manage Student (Amala Karthigayan):

User Interface Name or Number	Description	User Interface Layout
Coordinator Dashboard	Displayed one the coordinator successfully logs into the system. It provides access to various navigations such as Manage Student Registration, Manage Supervisee Quota and View Feedback.	Refer Appendix B.2
Student Registration Page	• Displays the existing batches of students registered for the FYP 1 subject. • <<Register New Batch>> button: Display the add new batch form.	
Add New Batch Form	Allows the coordinator to create a new batch by filling in the batch details.	
Student List Page	• <<Import Document>> button: Allows coordinator to upload the CSV or xlsx document containing list of pre-registered student details. • <<Generate Credential>> button: System creates unique UserID and password for every pre-registered student. • Display registered student list with their generated unique UserID and password.	
Supervisee Quota	• Displays a list of active supervisors. • Allows the Coordinator to manage (set & update) the supervisee quotas for each lecturer.	
View Feedback	Allows the Coordinator to view feedbacks and mark the feedbacks that has been viewed from students about the system.	

User Interface Description for Manage Final Year Project (MIRZA RUSYAIDI BIN RAHAMAT):

User Interface Name or Number	Description	User Interface Layout
Lecturer Dashboard	Displayed the Lecturer Dashboard that have three main function which is Registered Supervisee, FYP Interest Topic and Supervisee Request Pending.	Refer Appendix B.3
Supervisee Details	Displayed all the details of supervisee which is name, matric ID and FYP title.	
FYP Interest Topic	Displayed all the FYP interest topic. At this page, lecturer can edit, view and delete the topic.	
Supervisee Request Pending	Displayed all the pending request by supervisee.	
Supervisor Profile	Displayed all the lecturer info which is name, staff id, email, phone number and expertise.	
Book Appointment	Displayed the lecturer details and have book appointment form if the student want to make a appointment with the selected lecturer.	

User Interface Description for Manage Supervisor Hunting (NABILAH BINTI MOHAMMAD SHAHIDAN):

User Interface Name or Number	Description	User Interface Layout
Student Dashboard	Displayed the Student Dashboard that successfully logs into the system. It allows the student to search, sort the research group interests and register a supervisor for their Final Year Project (FYP). Students can search for prospective supervisors and sort based on research group regarding their interest's topic for their Final Year Project (FYP) and submit requests to register as a supervisee.	Refer Appendix B.4
Student Search	• Searches by lecturer name as their prospect supervisor. • Display list of supervisors.	
Student Sort	Allows the student to sort their interest topic Final Year Project based on research group.	
Supervisor Registration Page	<<Submit>> button: Allows the student to register their chosen supervisor.	
Lecturer Dashboard	Display the Lecturer Dashboard that successfully logs into the system. It allows the lecturers to approve or reject these requests.	
Supervisee Request Approval Form (Pending) Page	Display the list of supervisee request pending.	
Supervisee Request Approval Form (Reject) Page & Supervisee Request Approval Form (Approved)	<<Status>> dropdown button: Allows the lecturer either to approve or reject the request supervisor application.	

3.2.2 Hardware Interface

Not applicable.

3.2.3 Software Interface

Table 3.5 Software Interface

Application	API interface and Descriptions
Email	**SMTP (Simple Mail Transfer Protocol)**: SMTP is the standard protocol for sending emails across the Internet. SMTP servers are used to send messages from an email client to an email server and to relay messages between email servers. SMTP uses port 25 by default, with encryption protocols like TLS (on port 587) to secure the communication.

4 REQUIREMENT TRACEABILITY

Table 4.1 Requirement Traceability for Use Case Manage Coordinator Activities (NUR ASYIKIN)

Requirement ID	Traceability ID	Description	Interface ID
Req101_FYPSHA_2024	ReqTrac101_FYPSHA_2024	<<Add schedule>> button The coordinator able to add schedule for announcement.	Refer Appendix B.1
Req102_FYPSHA_2024	ReqTrac102_FYPSHA_2024	<<Publish>> The coordinator able to add announcement to display on the website for all to see.	
Req103_FYPSHA_2024	ReqTrac103_FYPSHA_2024	<<Generate Report >> button. The coordinator can monitor the system activities by generating reports. The coordinator can click the <<save>> button to save the generated report.	

Table 4.2 Requirement Traceability for Use Case Manage Student (AMALA)

Requirement ID	Traceability ID	Description	Interface ID
Req201_FYPSHA_2024	ReqTrac201_FYPSHA_2024	Coordinator registers students who have pre-registered for the FYP 1 course, providing them with system access. <<Register Student>> button <<Add New Batch>> button <<Import Document>> button <<Generate Credentials>> button	Refer Appendix B.2

Req202_FYPSHA_2024	ReqTrac202_FYPSHA_2024	Coordinator able to set and update the quota of supervisee for each lecturer. <<Manage Quota>> button	
Req203_FYPSHA_2024	ReqTrac203_FYPSHA_2024	Each lecturer is not allowed to receive more than their assigned quota for the students.	
Req204_FYPSHA_2024	ReqTrac204_FYPSHA_2024	System should be able to update in real time the available quota for each lecturer	
Req205_FYPSHA_2024	ReqTrac205_FYPSHA_2024	Any suggestion(s) function to increase the efficiency of this application are most welcome. <<View Feedback>> button The Coordinator clicks the <<View Feedback>> button to view feedback submitted by students.	
Req206_FYPSHA_2024	ReqTrac206_FYPSHA_2024	Easy to use and require minimum experience to learn the system for first-time users. User Friendly Interface Design	

Table 4.3 Requirement Traceability for Use Case Manage Final Year Project (MIRZA RUSYAIDI BIN RAHAMAT)

Traceability ID	Traceability ID	Description	Interface ID
Req301_FYPSHA_2024	ReqTrac301_FYPSHA_2024	Fill in form <<FYP Interest Topic>> Lecturer allows to add FYP interest topic.	Refer Appendix B.3
Req302_FYPSHA_2024	ReqTrac302_FYPSHA_2024	Update Lecturer Profile form <<Expertise>> Lecturer allows to update their expertise at lecturer profile.	
Req303_FYPSHA_2024	ReqTrac303_FYPSHA_2024	Fill in appointment form <<Date, Time, Reason>> Student allows to make appointment with desired lecturer.	

Table 4.3 Requirement Traceability for Use Case Manage Supervisor Hunting (NABILAH BINTI MOHAMMAD SHAHIDAN)

Requirement ID	Traceability ID	Description	Interface ID
Req401_FYPSHA_2024	ReqTrac401_FYPSHA_2024	Fill in search box <<Lecturer Name>> The student allows to search for prospective supervisors by searching lecturer name.	Refer Appendix B.4
Req402_FYPSHA_2024	ReqTrac402_FYPSHA_2024	Sorts drop down button <<Research Group>> Display <<List of Lecturer>>	

		The student sorts the research group interests for their Final Year Project (FYP).	
Req403_FYPSHA_2024	ReqTrac403_FYPSHA_2024	Fill in the register form <<Title, Brief Description>> Click <<Submit>> button. The student registers a lecturer as supervisor by submitting the register form.	
Req404_FYPSHA_2024	ReqTrac404_FYPSHA_2024	Approval Validation The lecturer can approve or reject these requests.	
Req405_FYPSHA_2024	ReqTrac405_FYPSHA_2024	Real-Time Quota Update Function The system should be able to update in real time the available quota for each lecturer.	
Req406_FYPSHA_2024	ReqTrac406_FYPSHA_2024	User Friendly Interface Design. Easy to use and require minimum experience to learn the system for first-time users.	

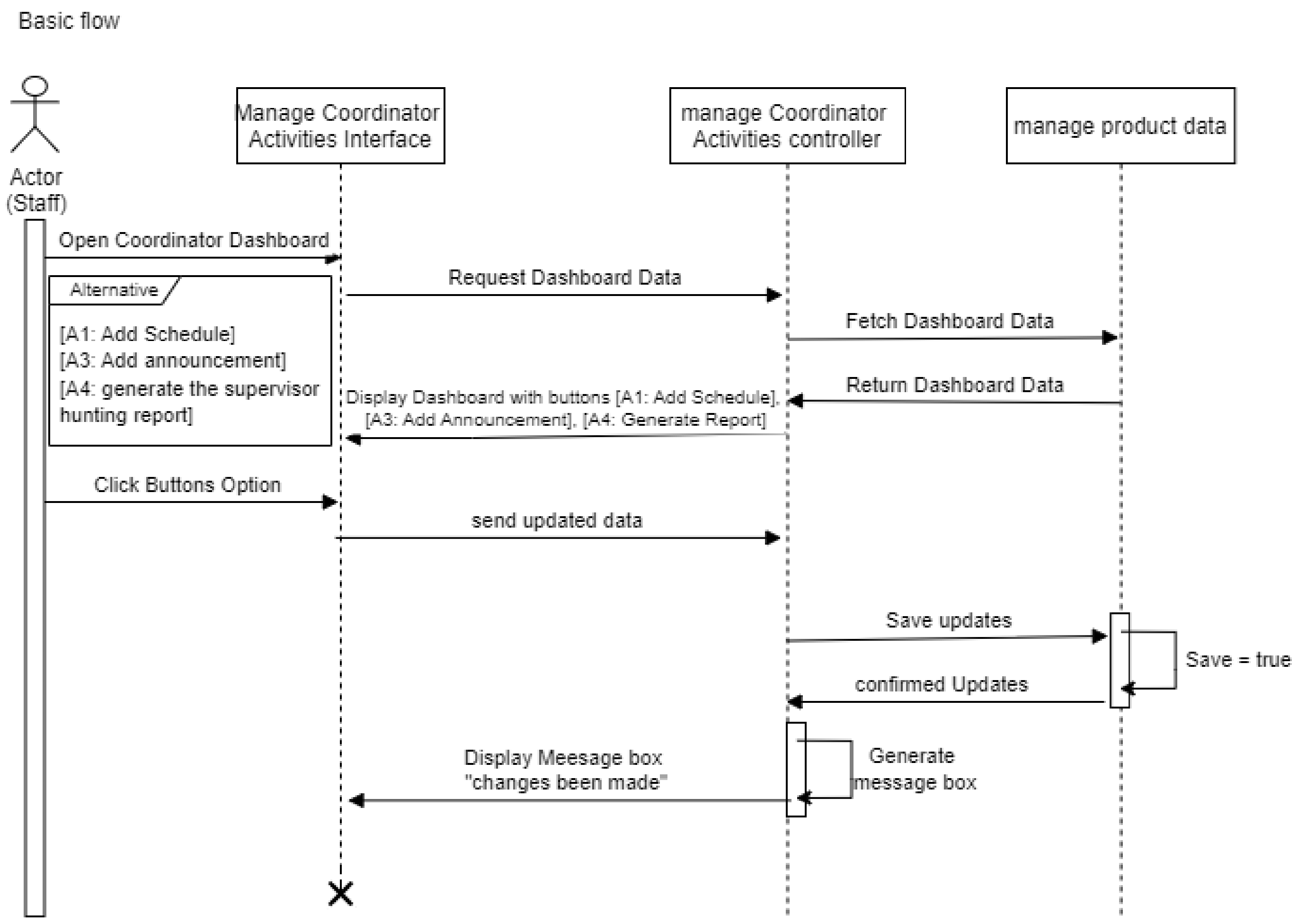
5 ACRONYMS AND ABBREVIATION

Table 5.1 List of Acronyms and Abbreviations

SRS	Software Requirement Specification
FYPSHA	Final Year Project Supervisor Hunting Application (System Name)
SRW	Software Requirement Workshop
HORG	Head of Research Group

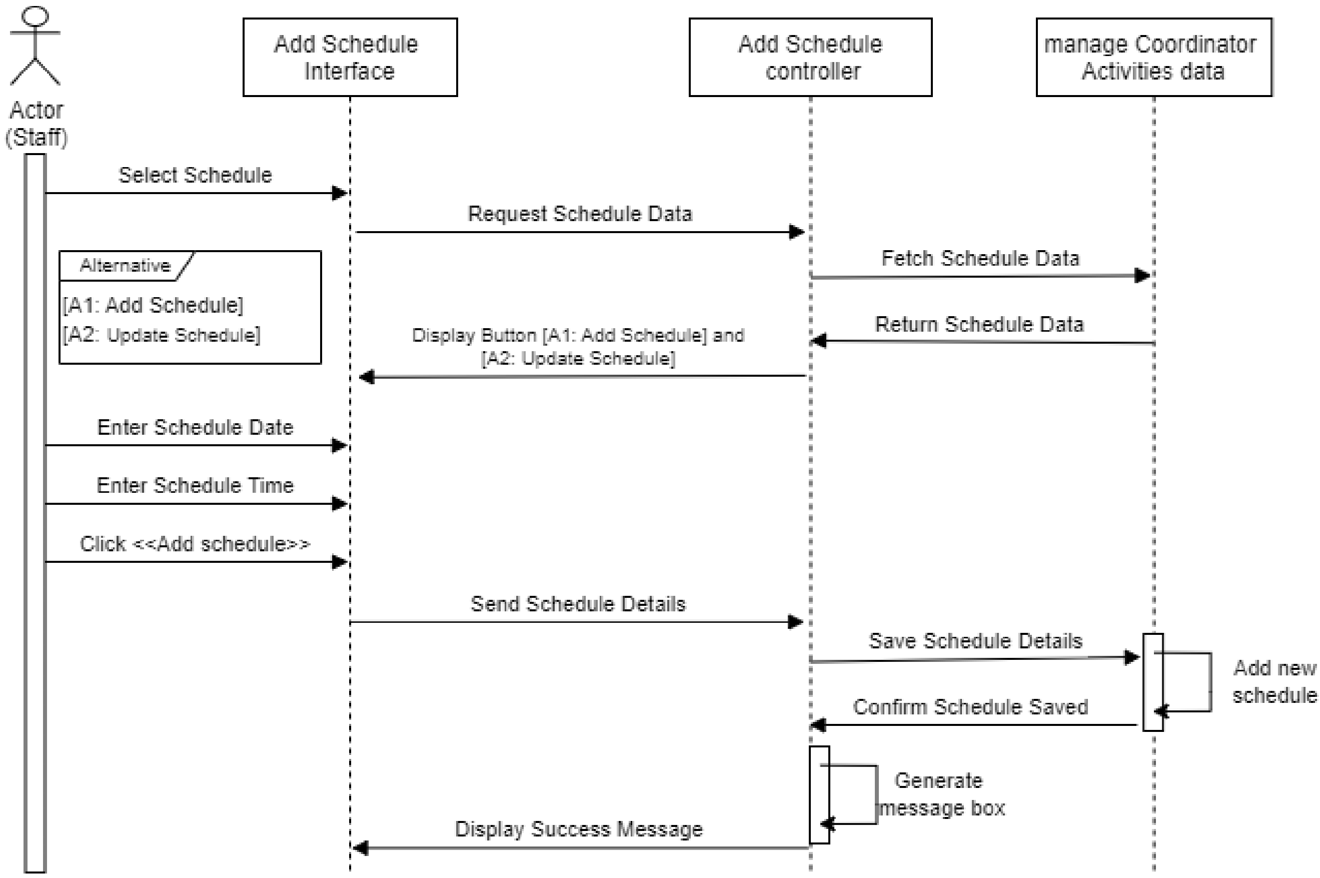
APPENDIX A
Sequence Diagram

1. Manage Coordinator Activities (NUR ASYIKIN)

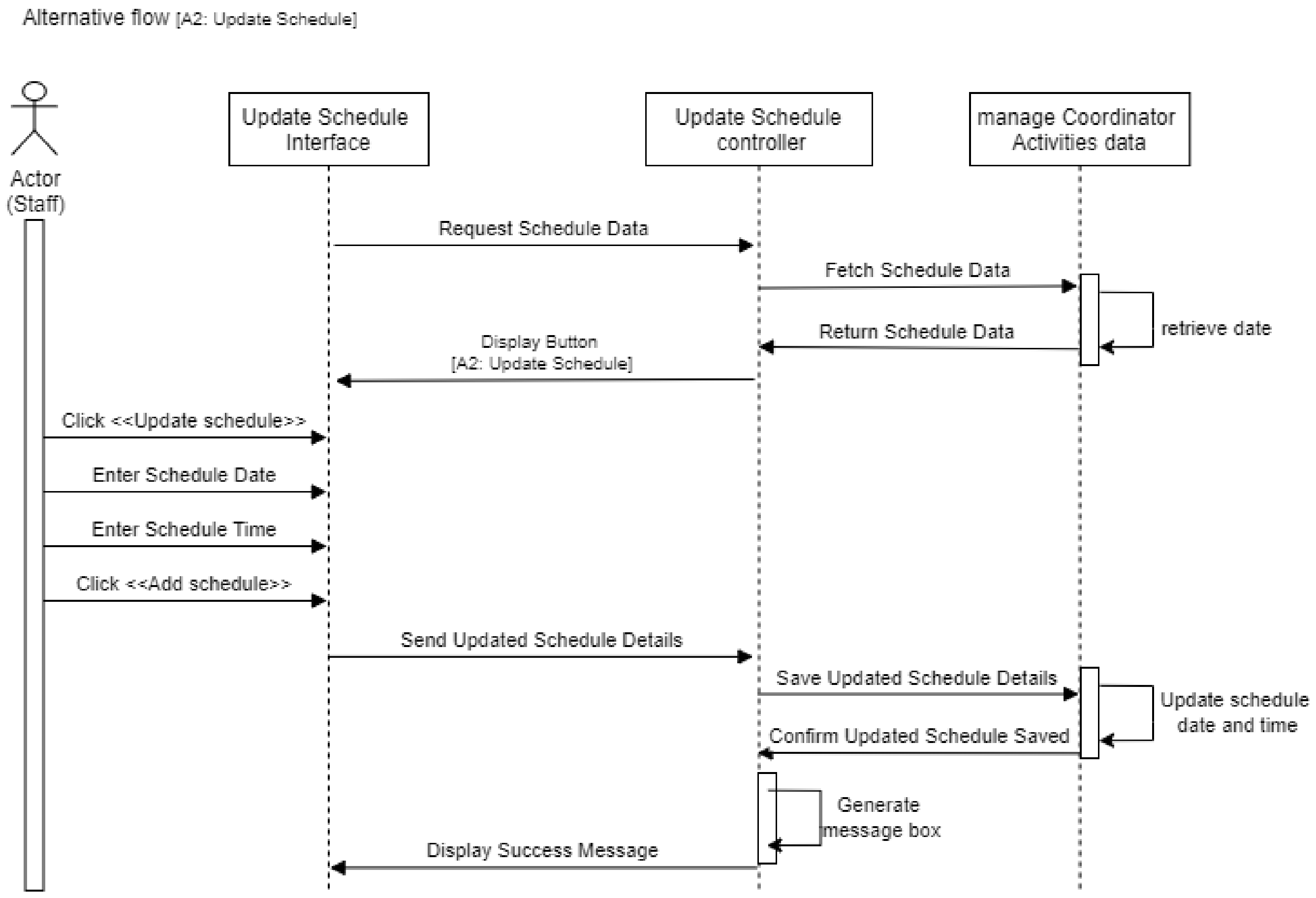


A-1.1: Sequence Diagram – Basic Flow

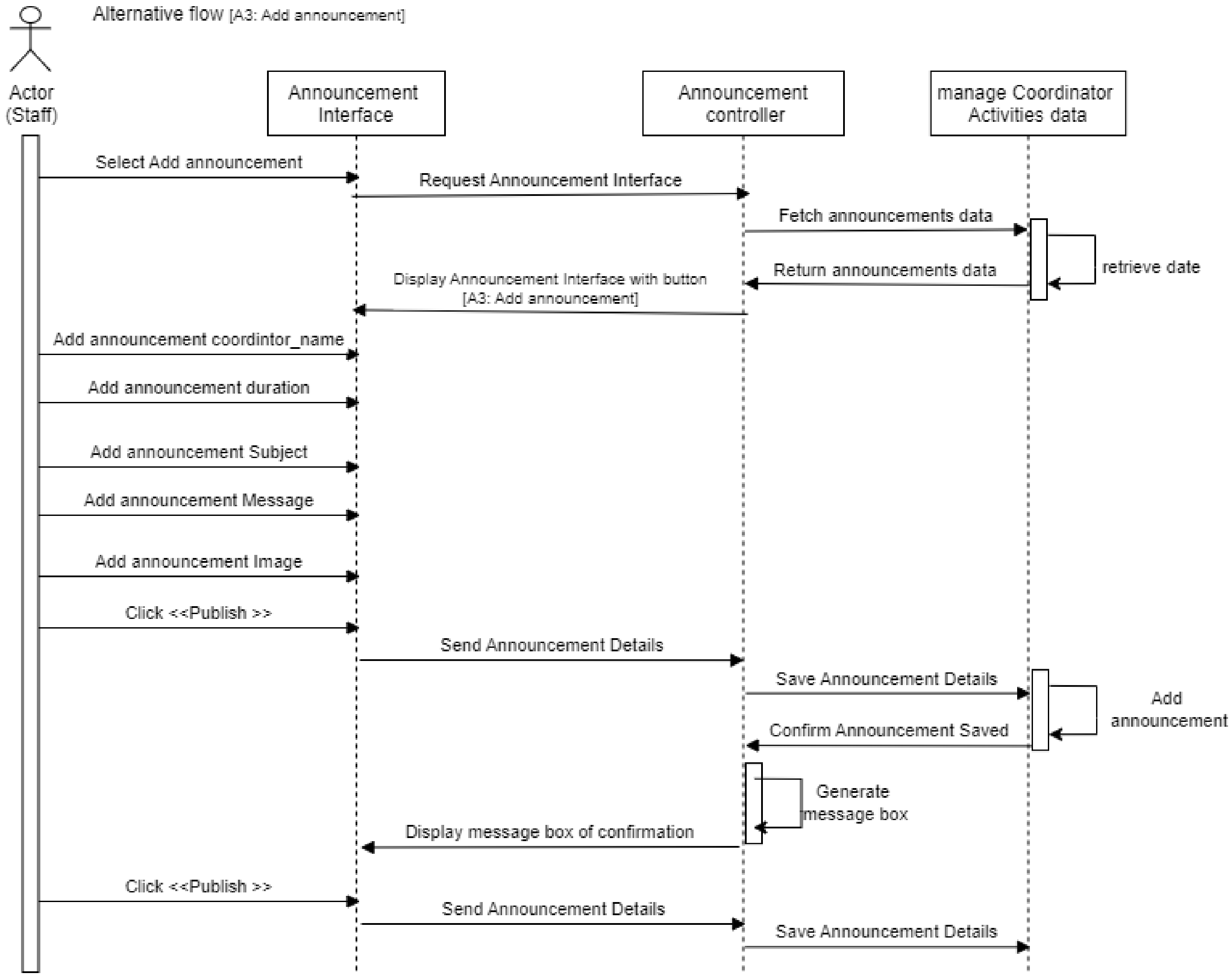
Alternative flow [A1: Add Schedule]



A-1.2.1: Sequence Diagram – Alternative Flow Add Schedule

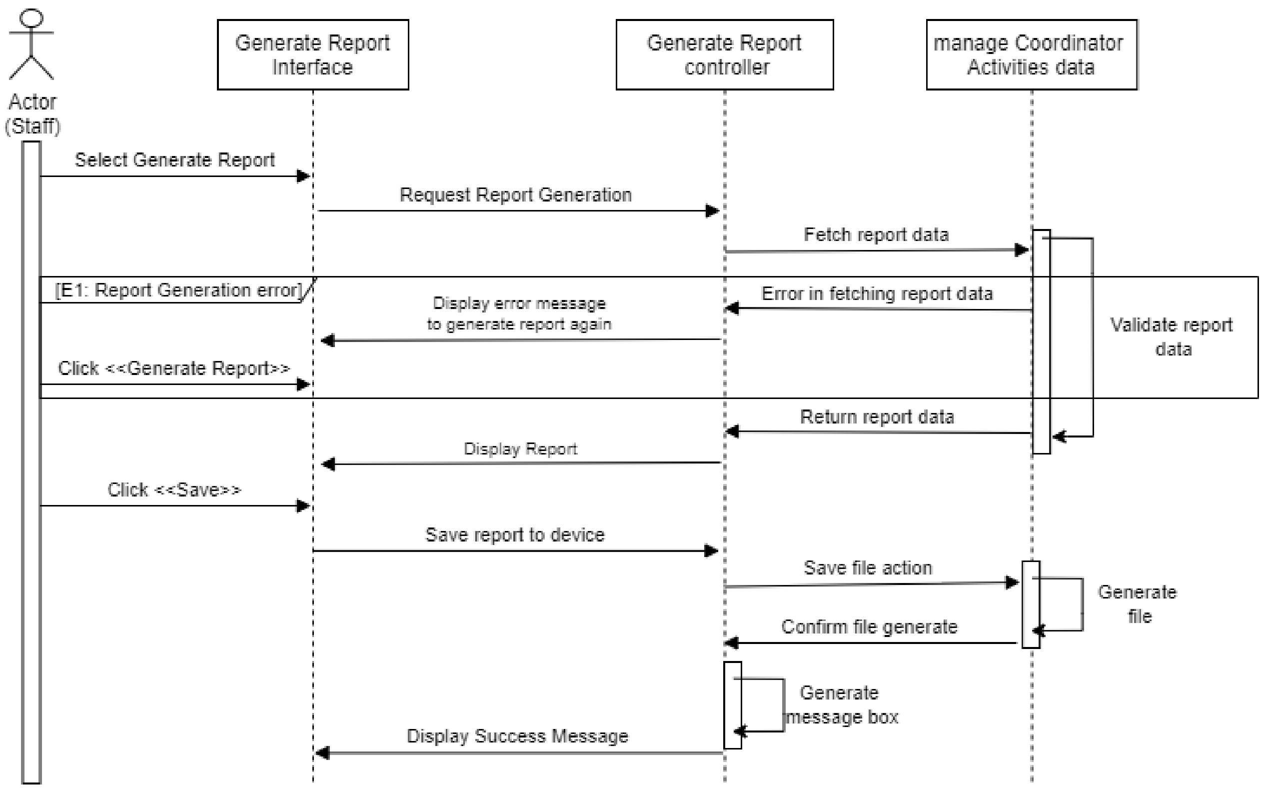


A-1.2.2: Sequence Diagram – Alternative Flow Update Schedule



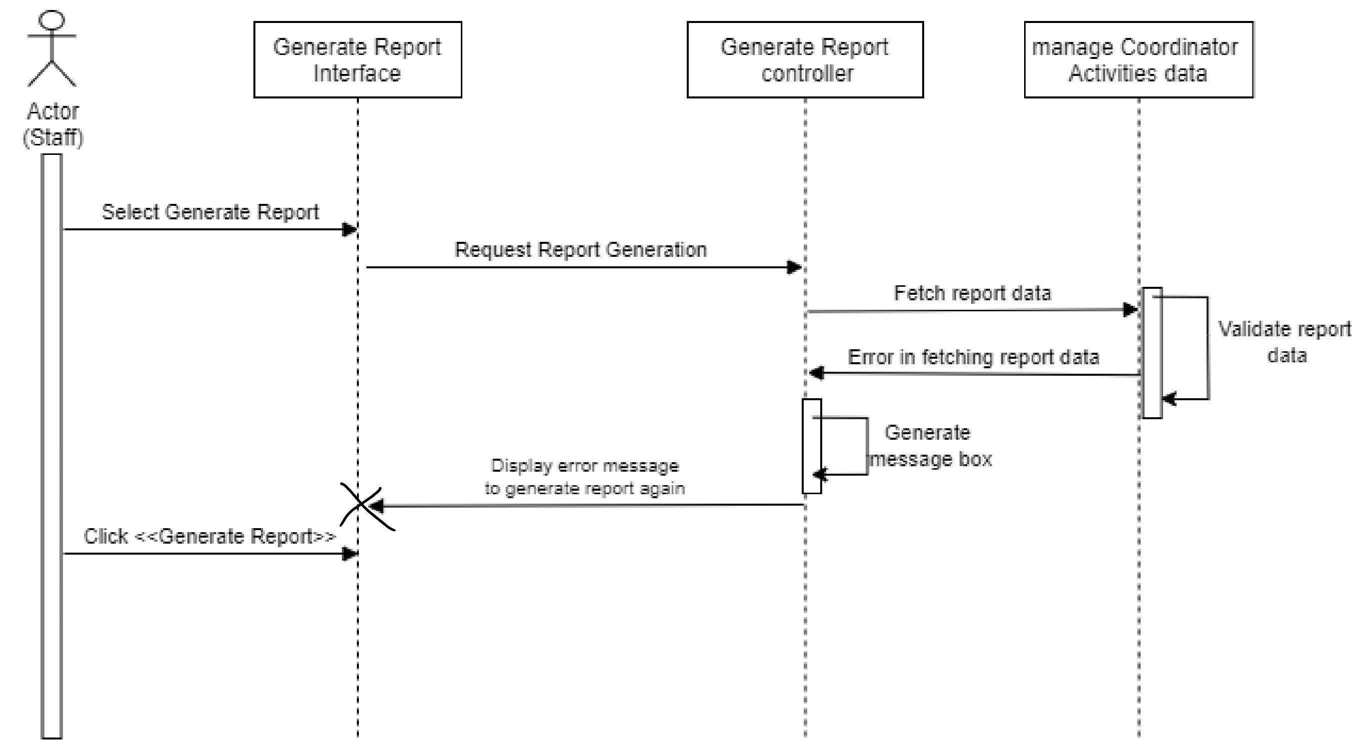
A-1.2.3: Sequence Diagram – Alternative Flow Add Announcement

Alternative flow [A4:Generate Report]



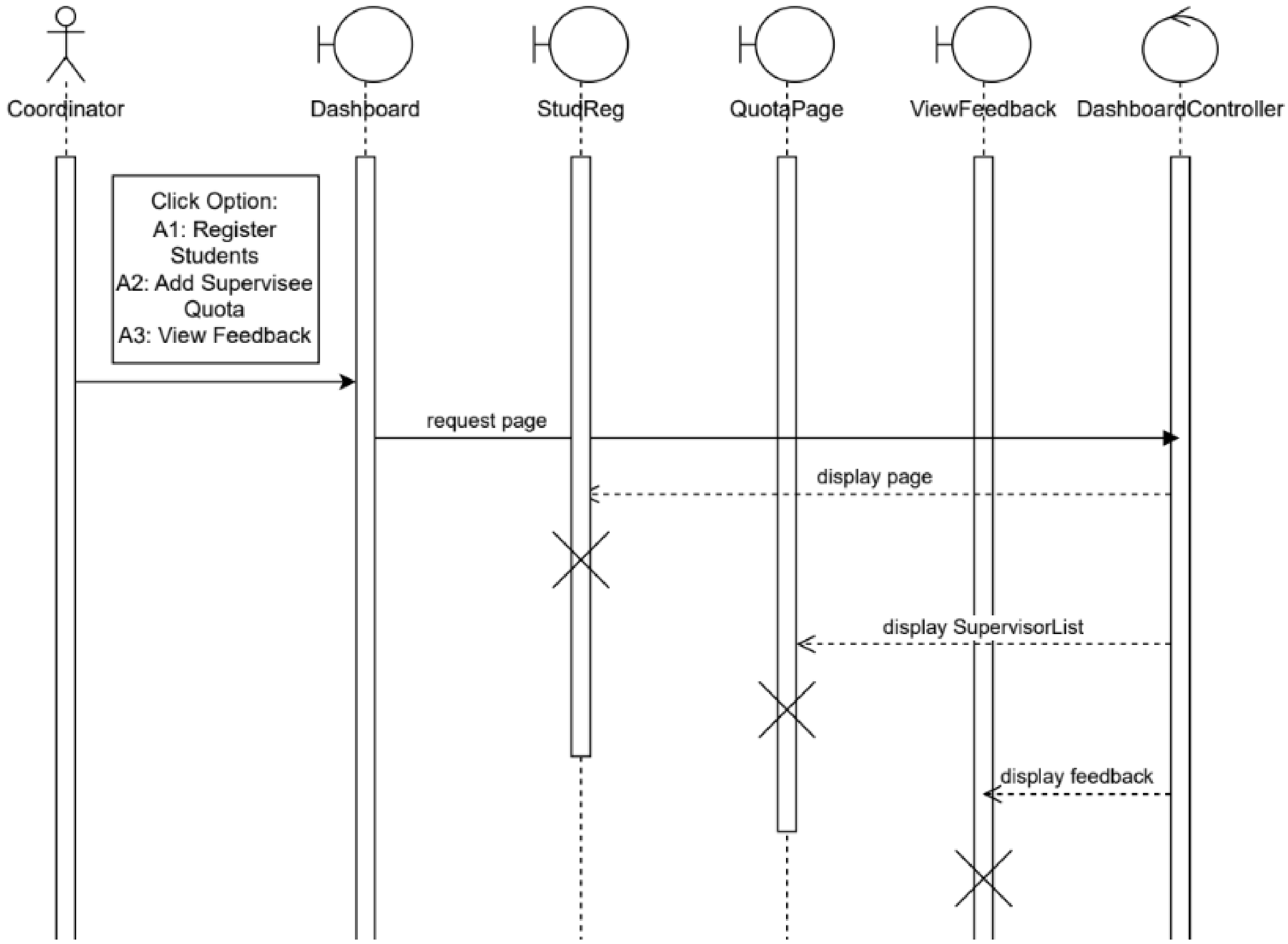
A-1.2.4: Sequence Diagram – Alternative Flow Generate Report

Exception flow [E1: Report Generation error]

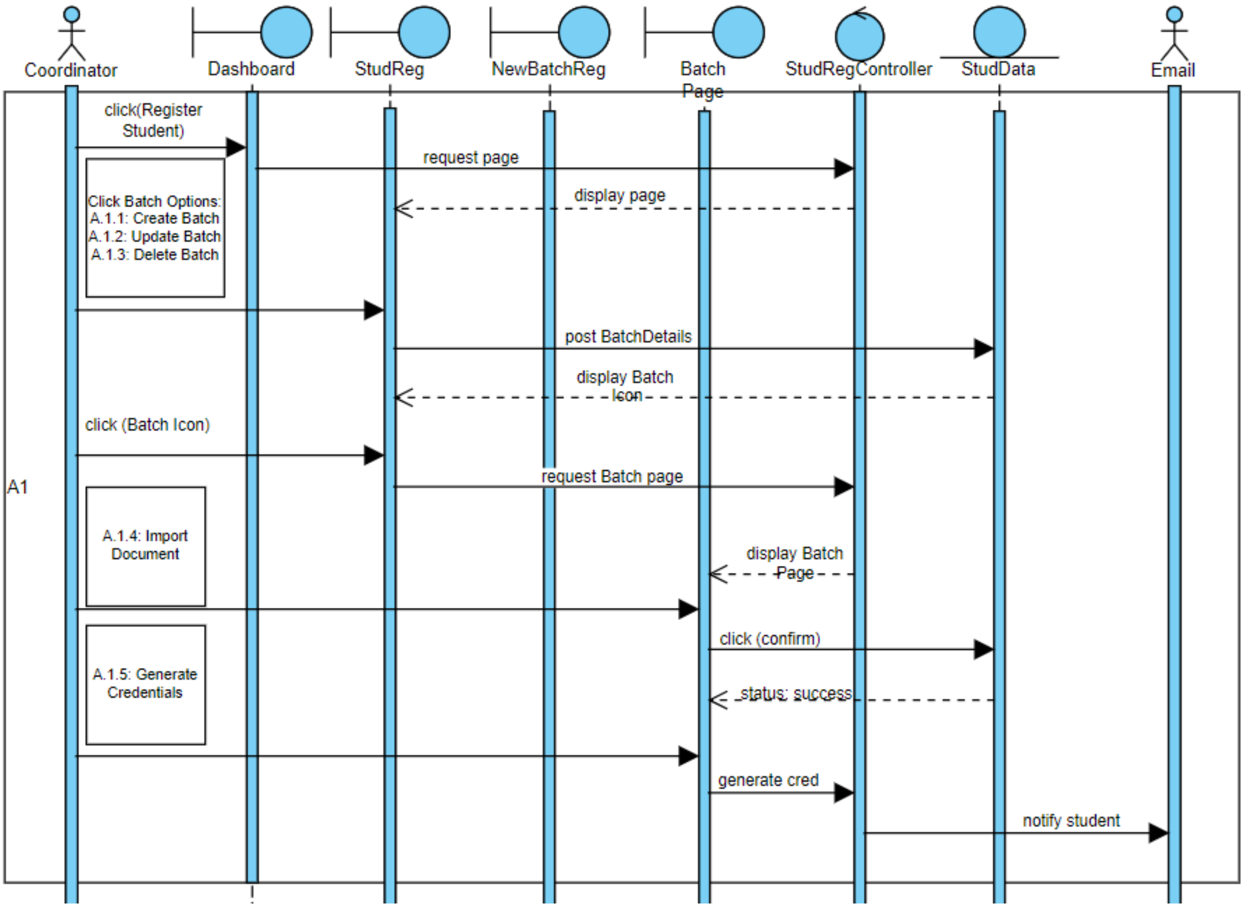


A-1.3: Sequence Diagram – Exception Flow Report Generation error

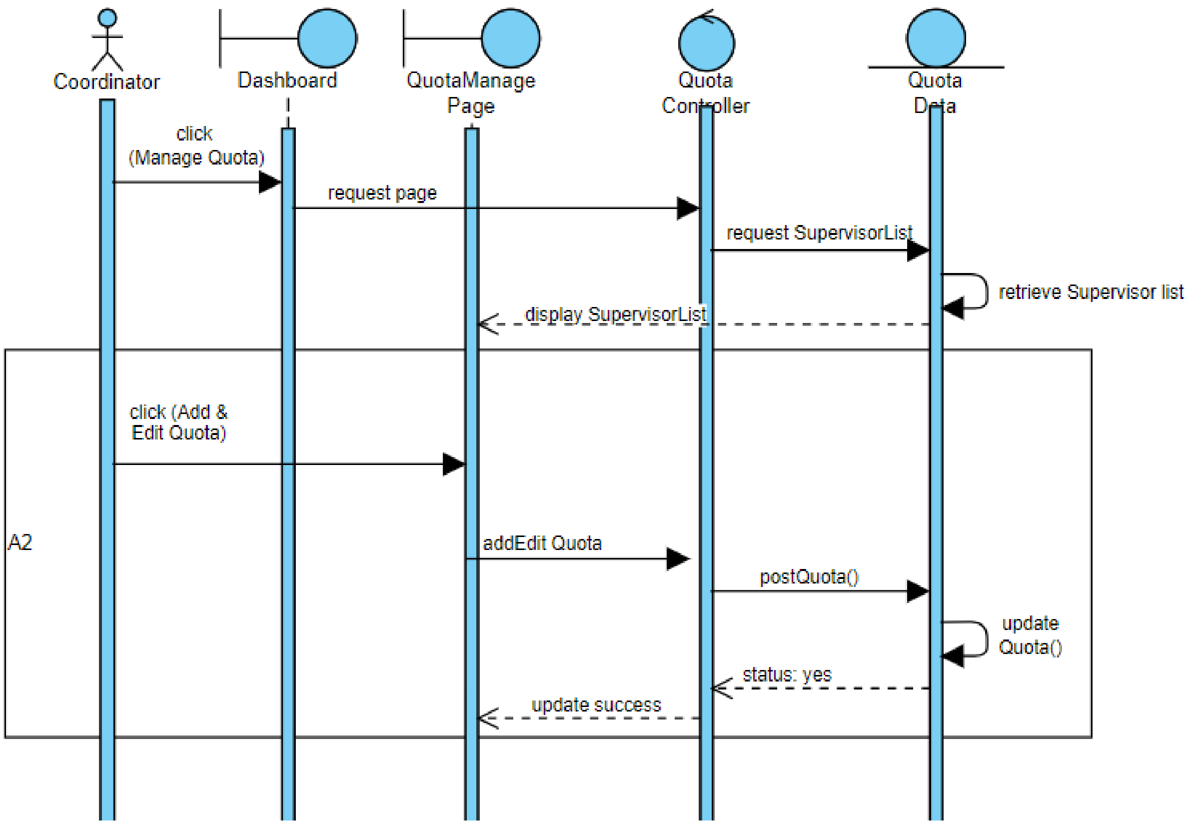
2. Manage Student (AMALA KARTHIGAYAN)



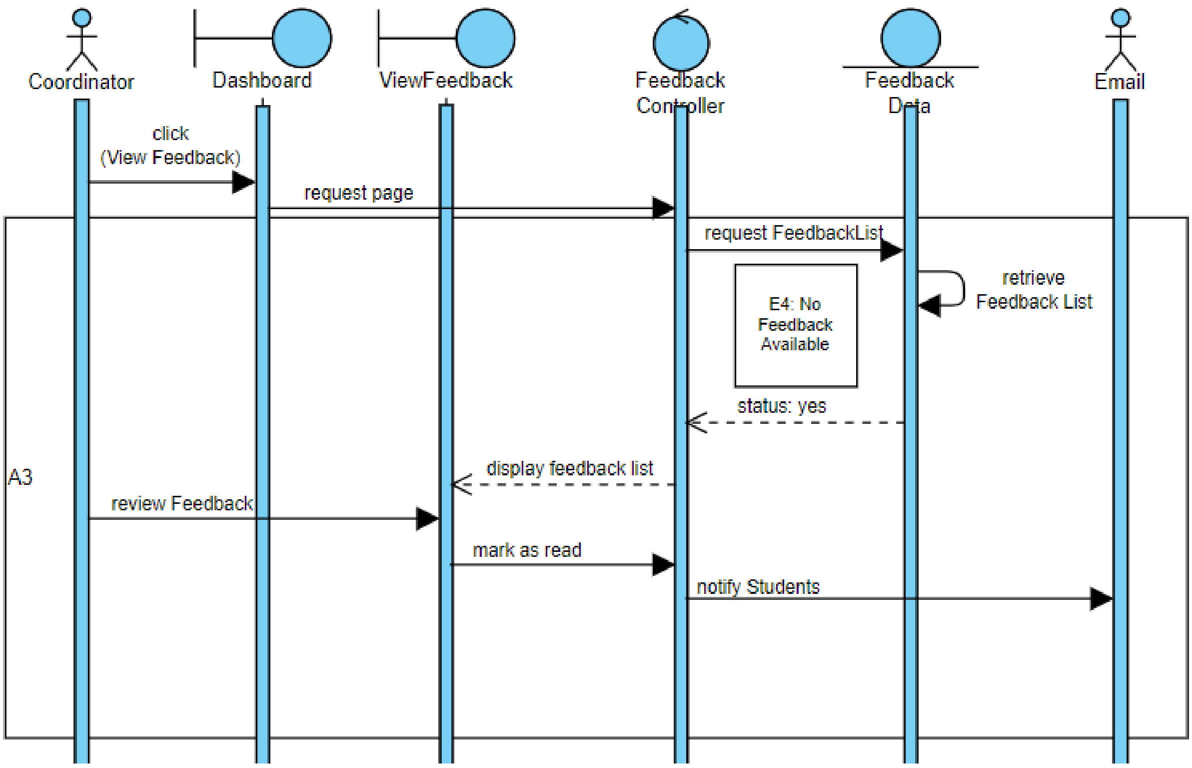
A-2.1: Sequence Diagram – Basic Flow



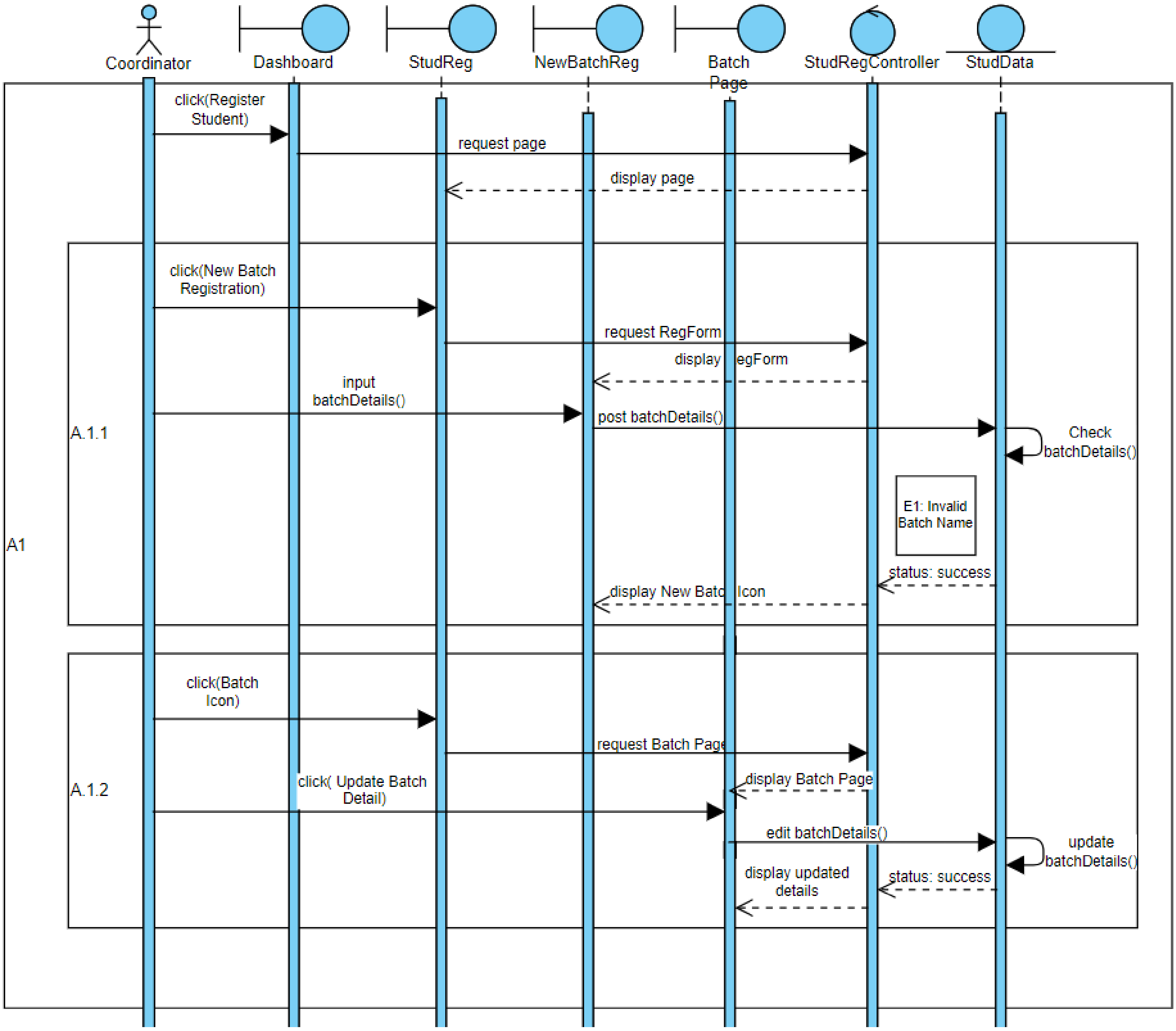
A-2.2.1: Sequence Diagram – Alternative Flow



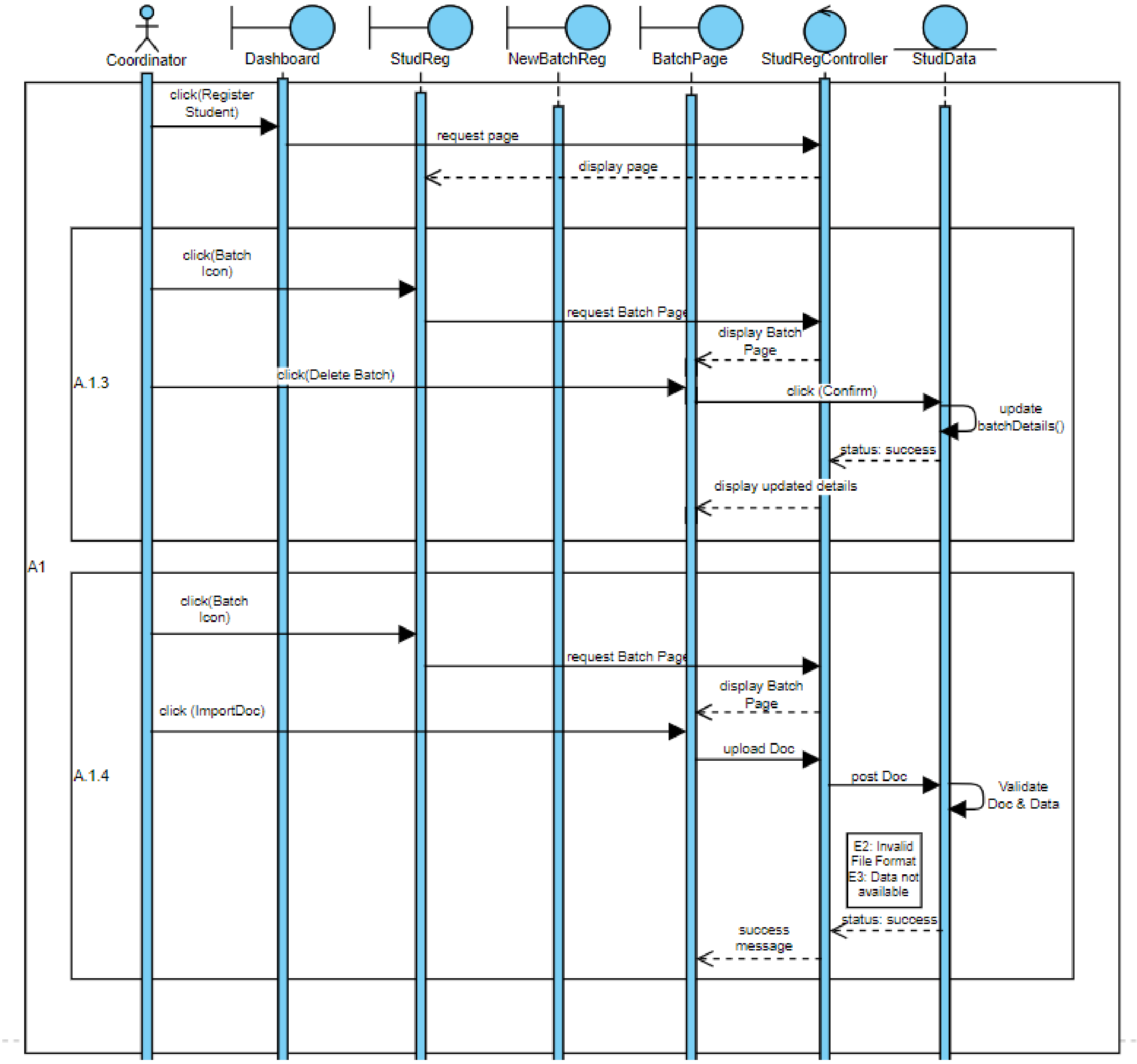
A-2.2.2: Sequence Diagram – Alternative Flow



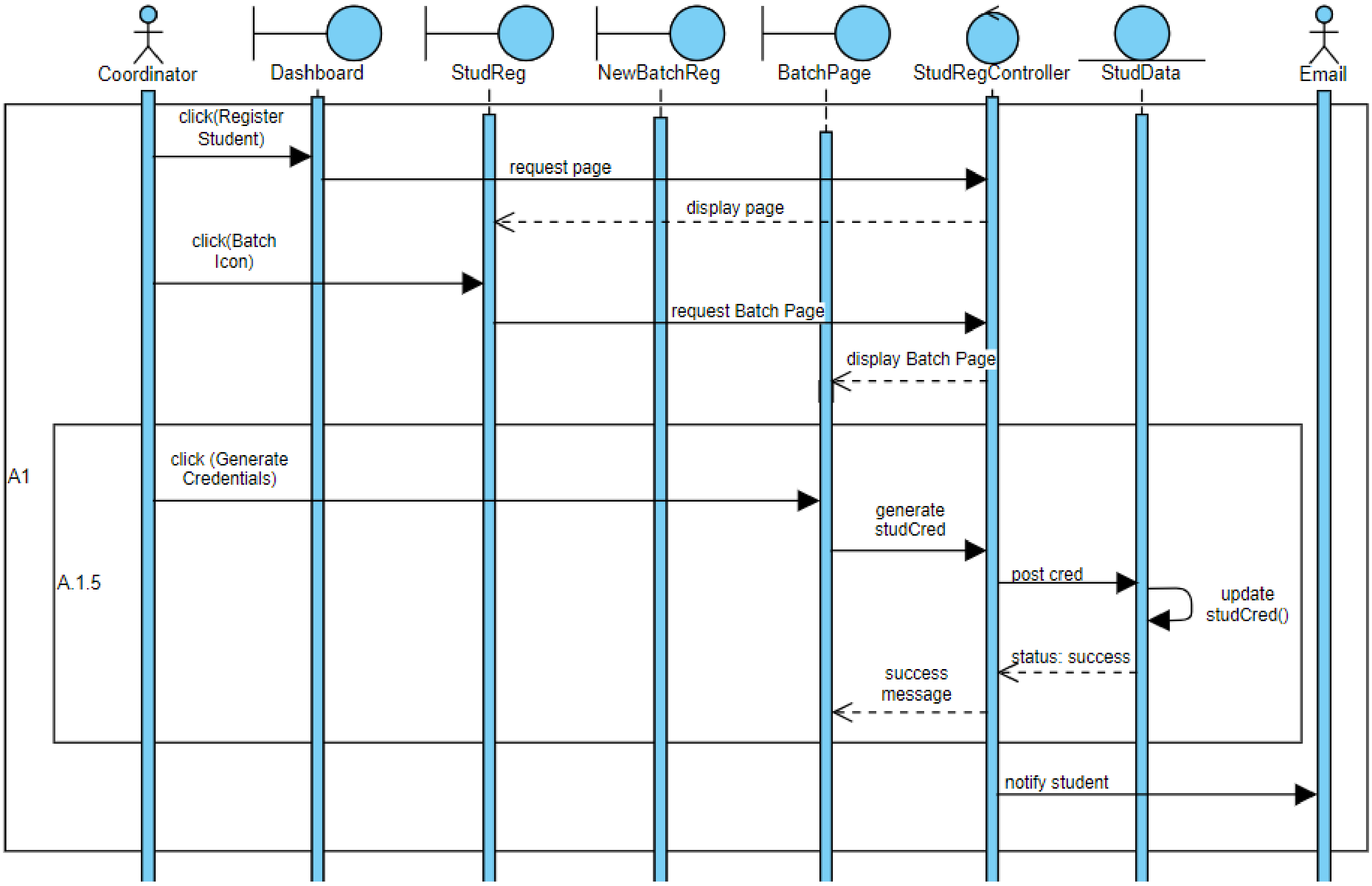
A-2.2.3: Sequence Diagram – Alternative Flow



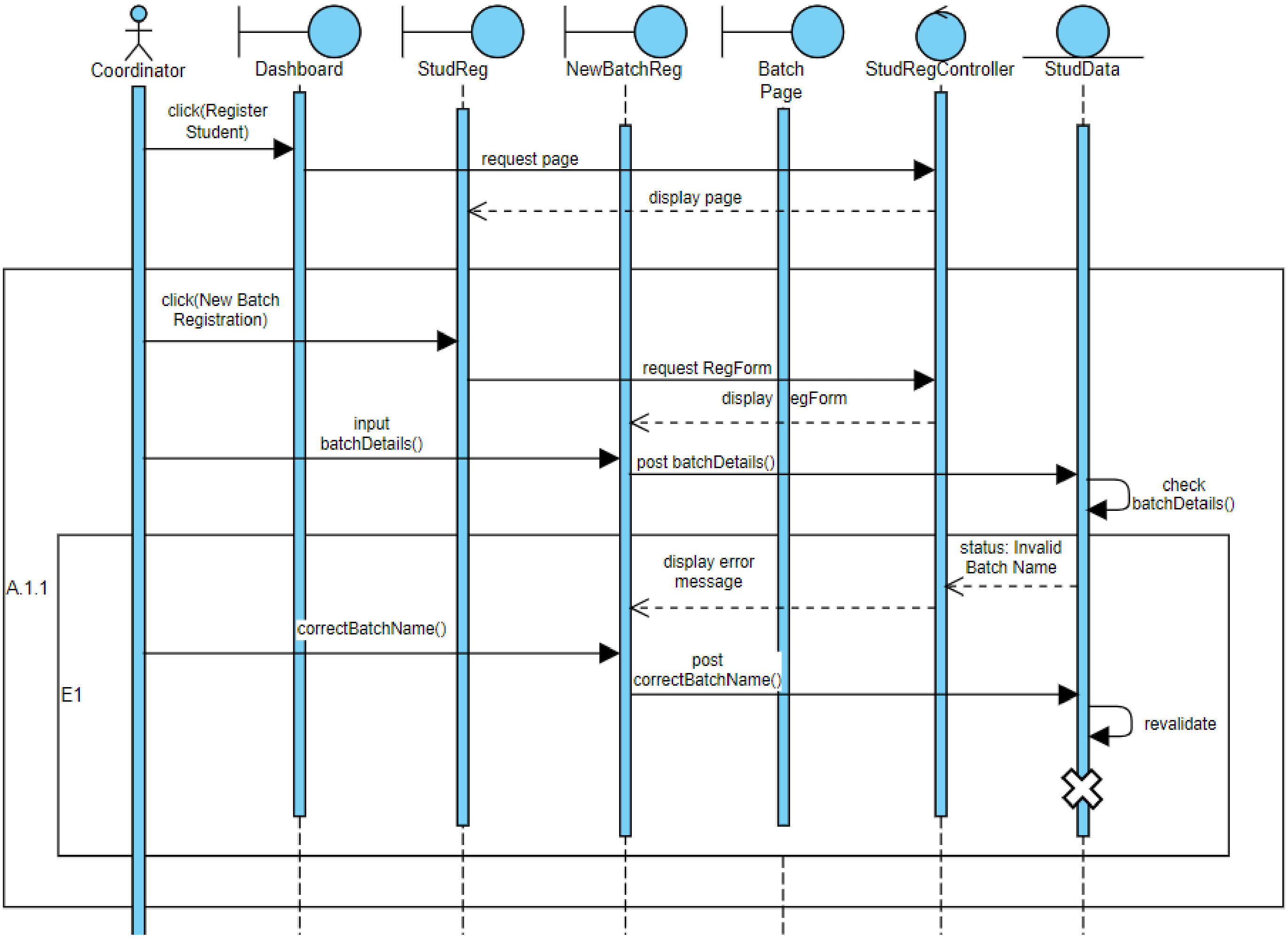
A-2.2.4: Sequence Diagram – Alternative Flow



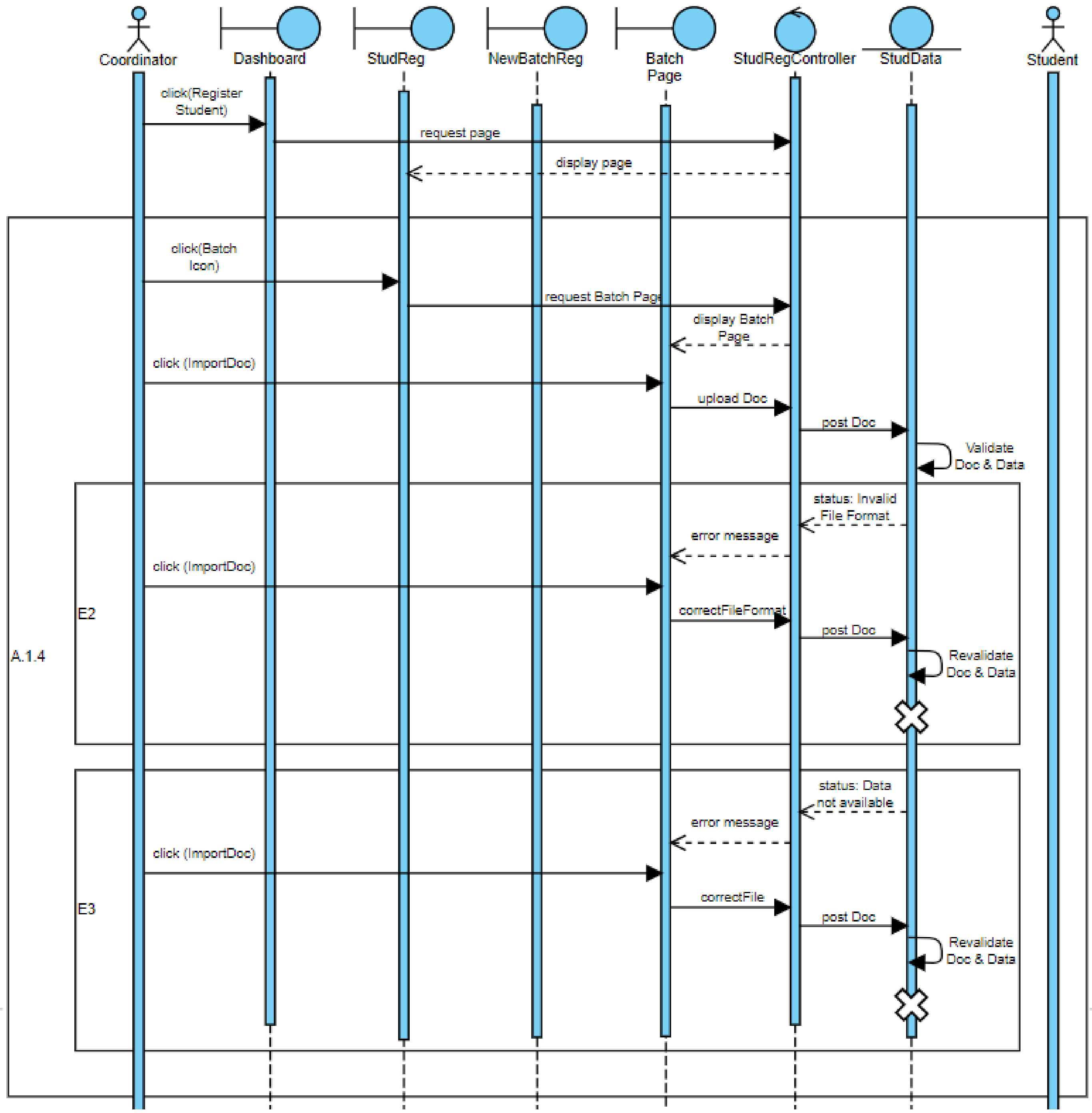
A-2.2.5: Sequence Diagram – Alternative Flow



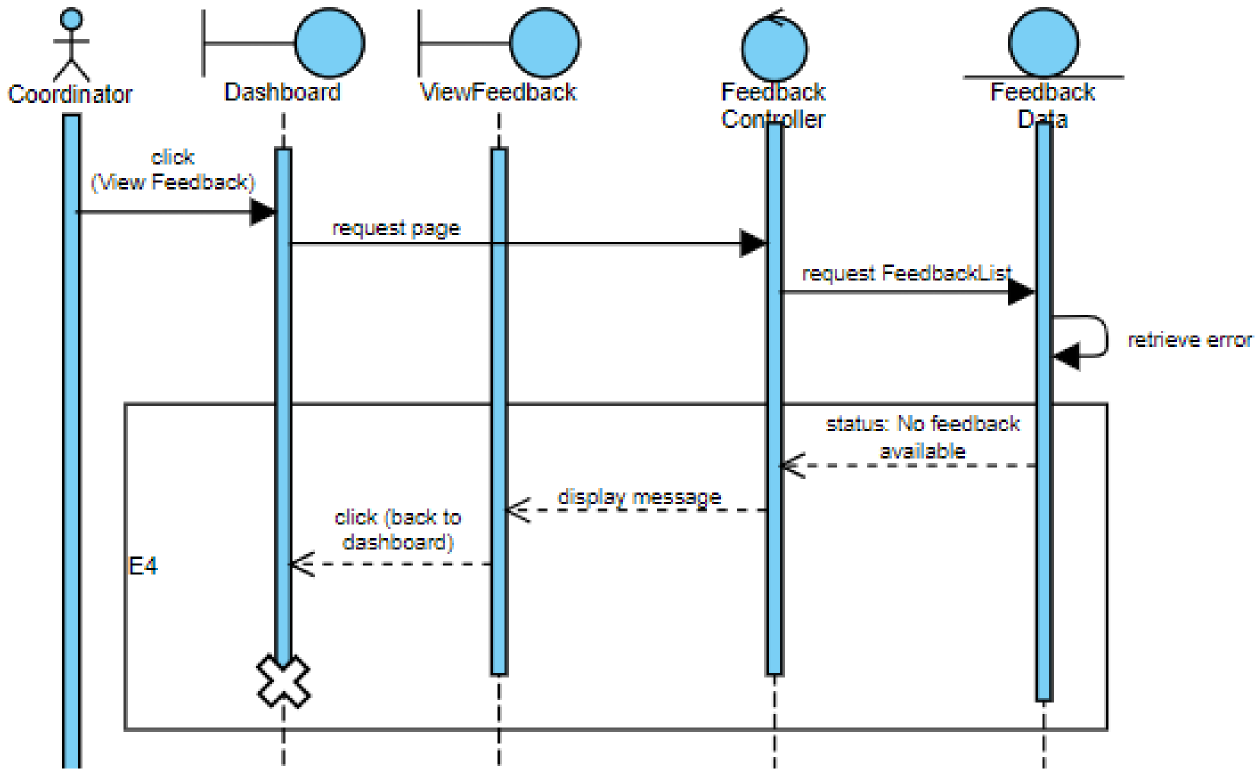
A-2.2.6: Sequence Diagram – Alternative Flow



A-2.3.1: Sequence Diagram – Exception Flow

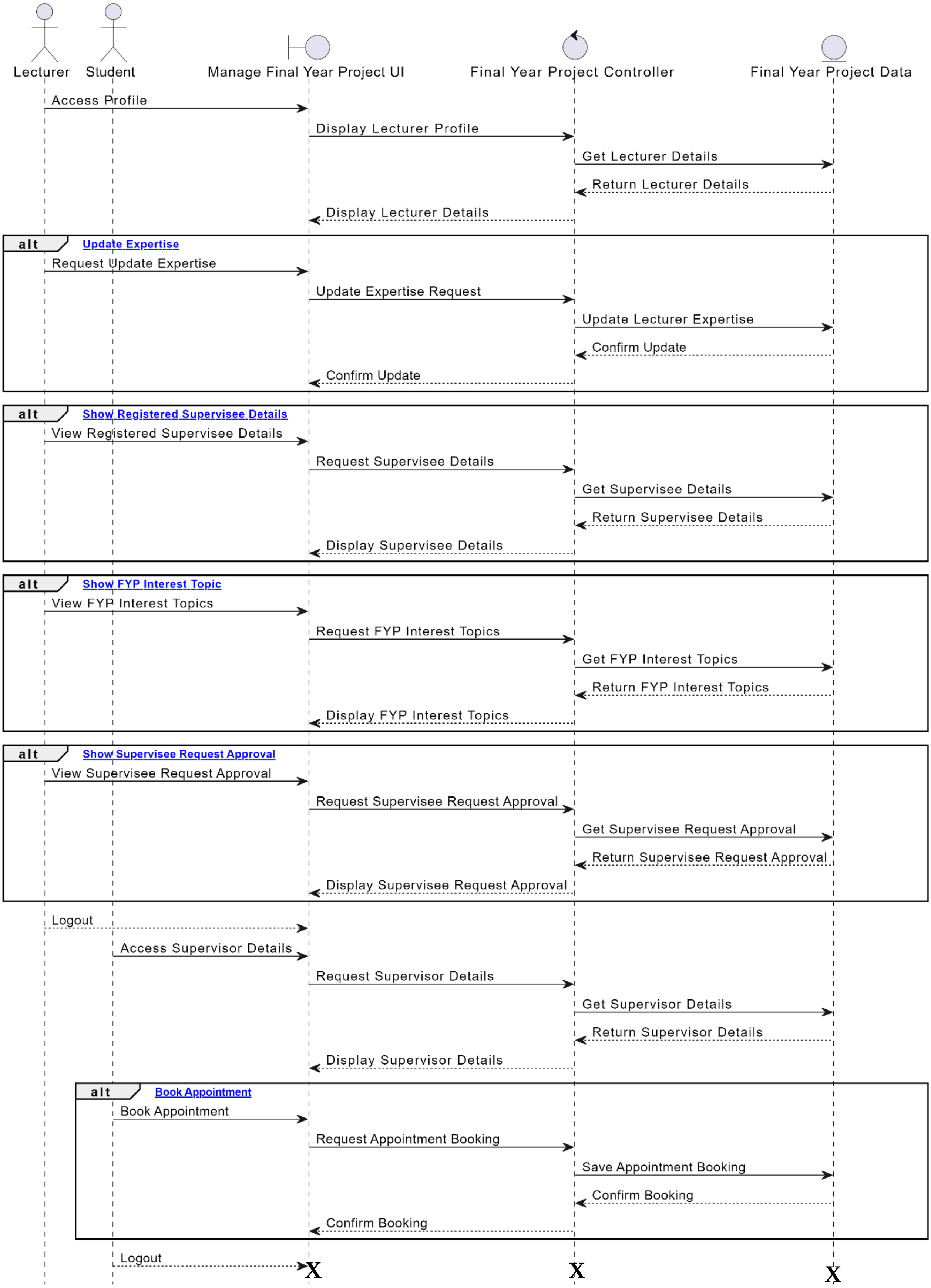


A-2.3.2: Sequence Diagram – Exception Flow

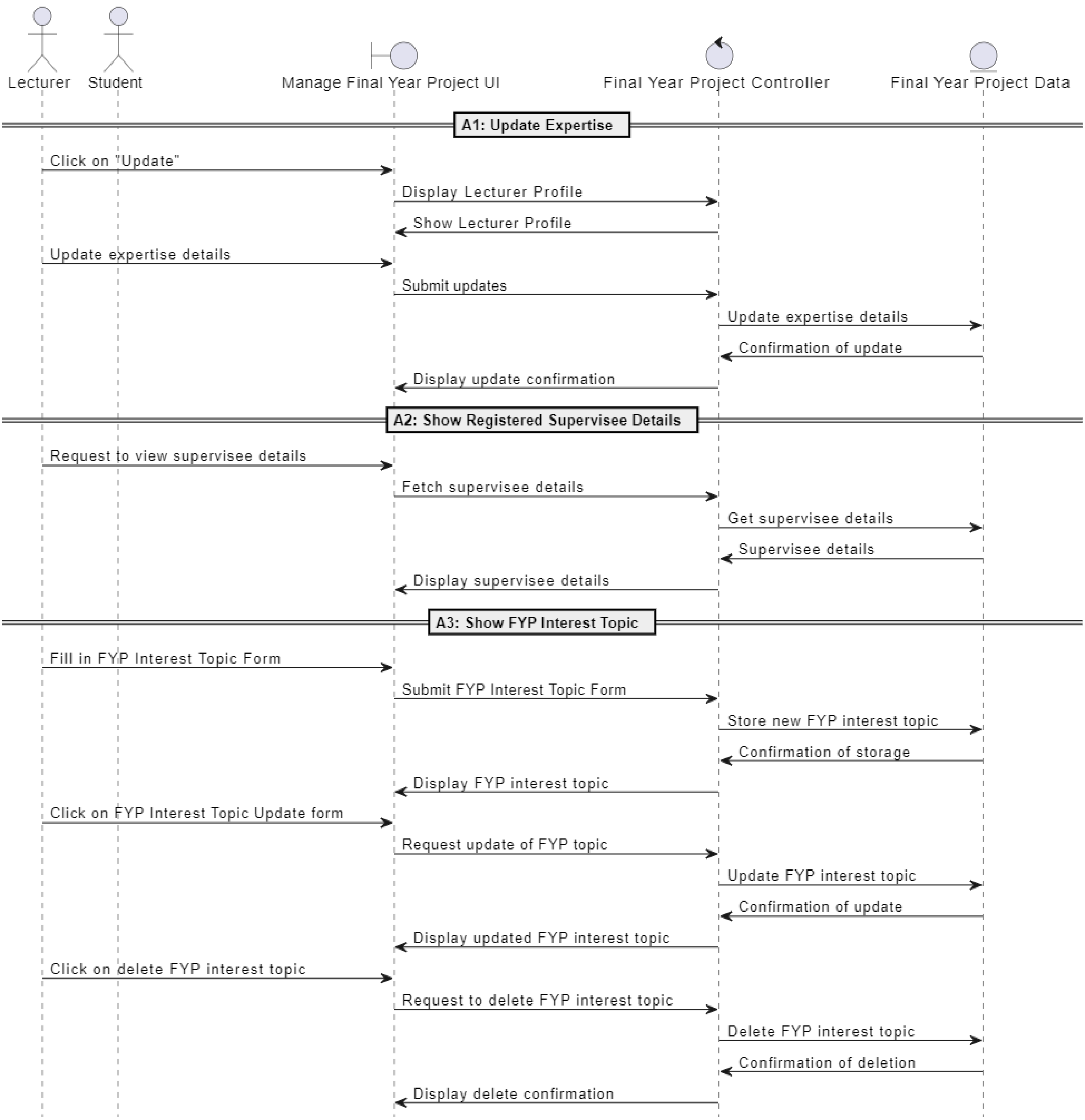


A-2.3.3: Sequence Diagram – Exception Flow

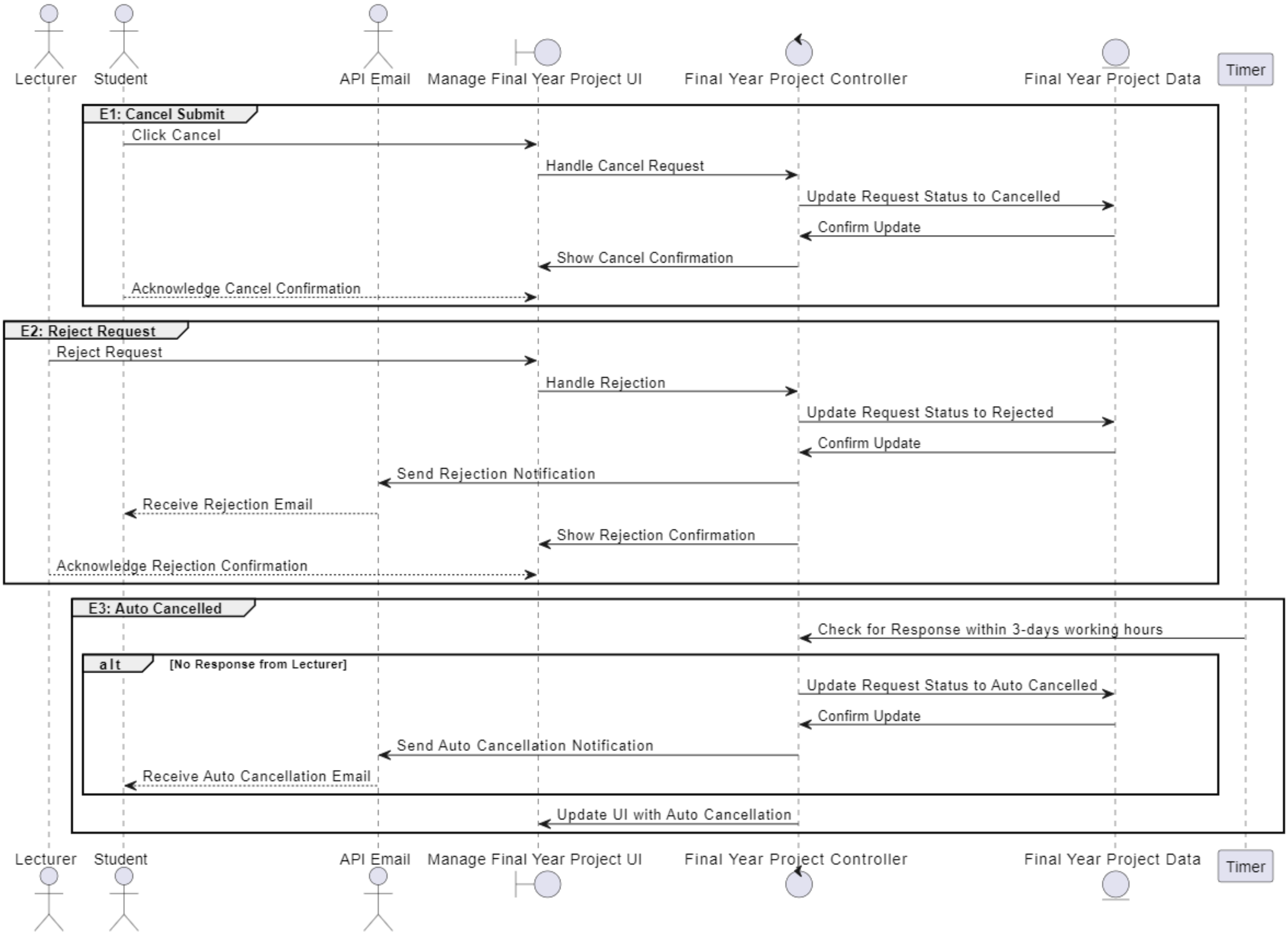
3. Manage Final Year Project (MIRZA RUSYAIDI)



A-3.1: Sequence Diagram – Basic Flow

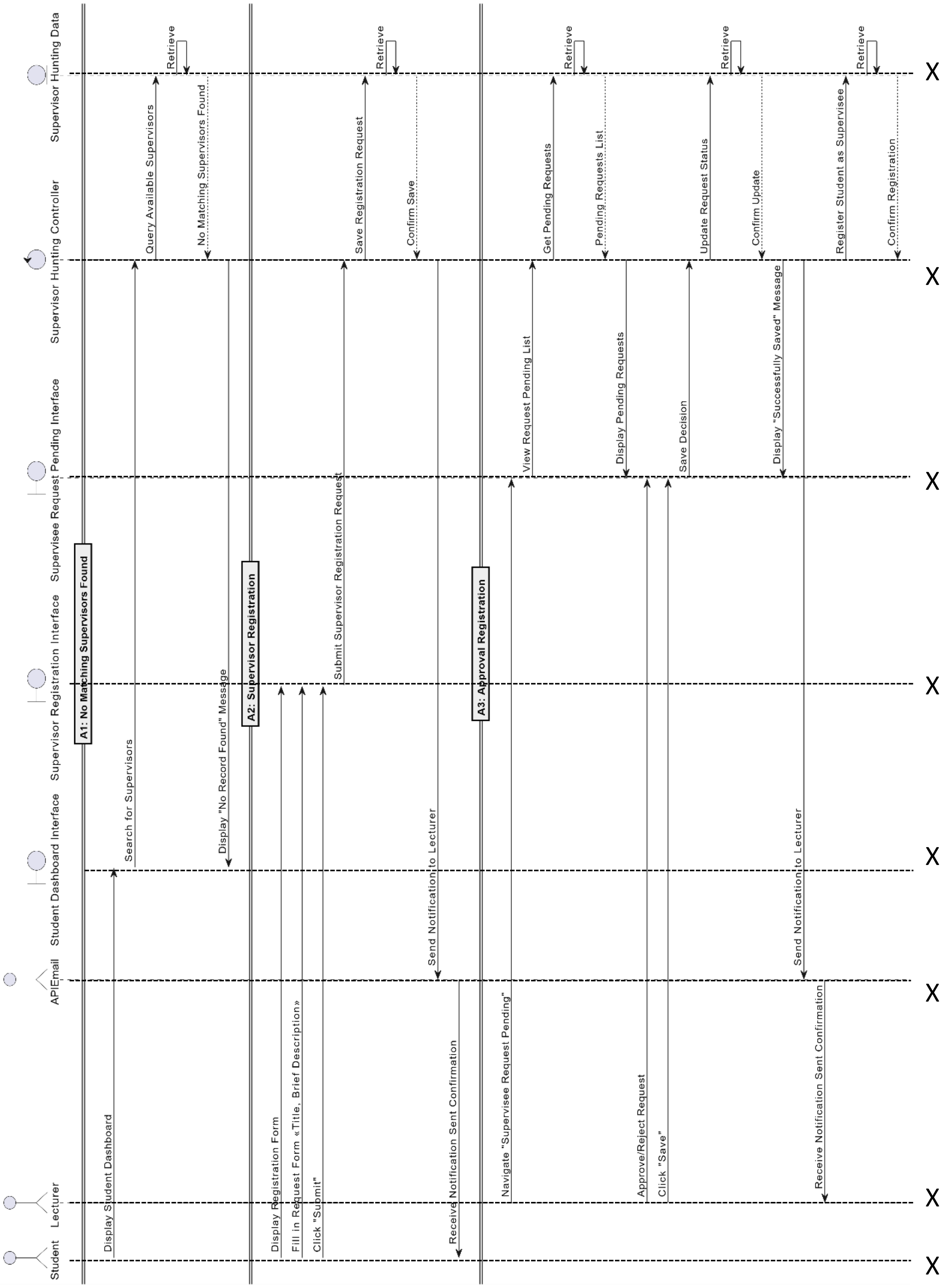

Lecturer
Student
Manage Final Year Project UI
Final Year Project Controller
Final Year Project Data
A1: Update Expertise
Click on "Update"
Display Lecturer Profile
Show Lecturer Profile
Update expertise details
Submit updates
Update expertise details
Confirmation of update
Display update confirmation
A2: Show Registered Supervisee Details
Request to view supervisee details
Fetch supervisee details
Get supervisee details
Supervisee details
Display supervisee details
A3: Show FYP Interest Topic
Fill in FYP Interest Topic Form
Submit FYP Interest Topic Form
Store new FYP interest topic
Confirmation of storage
Display FYP interest topic
Click on FYP Interest Topic Update form
Request update of FYP topic
Update FYP interest topic
Confirmation of update
Display updated FYP interest topic
Click on delete FYP interest topic
Request to delete FYP interest topic
Delete FYP interest topic
Confirmation of deletion
Display delete confirmation

A-3.2: Sequence Diagram – Alternative Flow

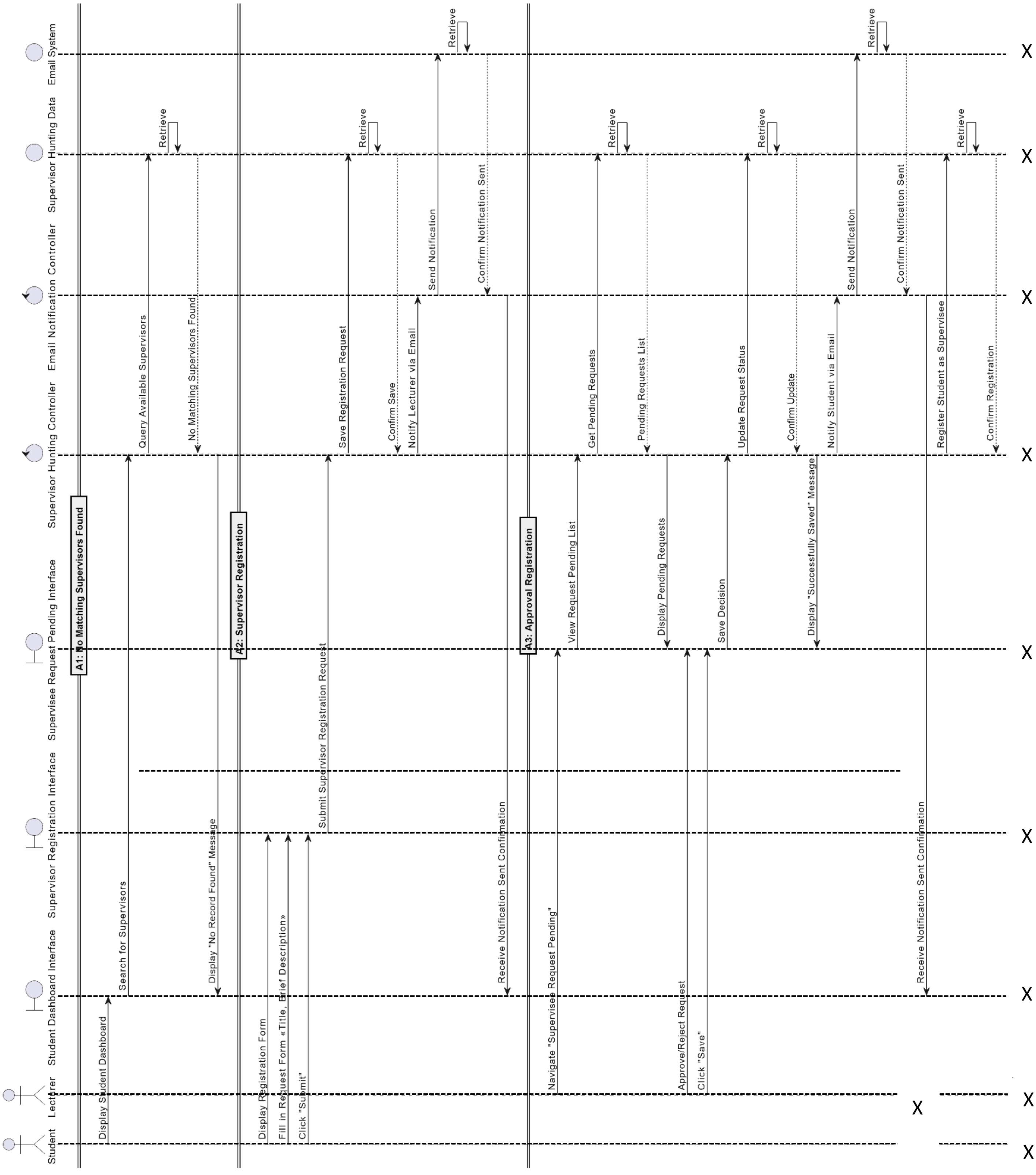


A-3.3: Sequence Diagram – Exception Flow

4. Manage Supervisor Hunting (NABILAH)



A-4.2: Sequence Diagram – Alternative Flow



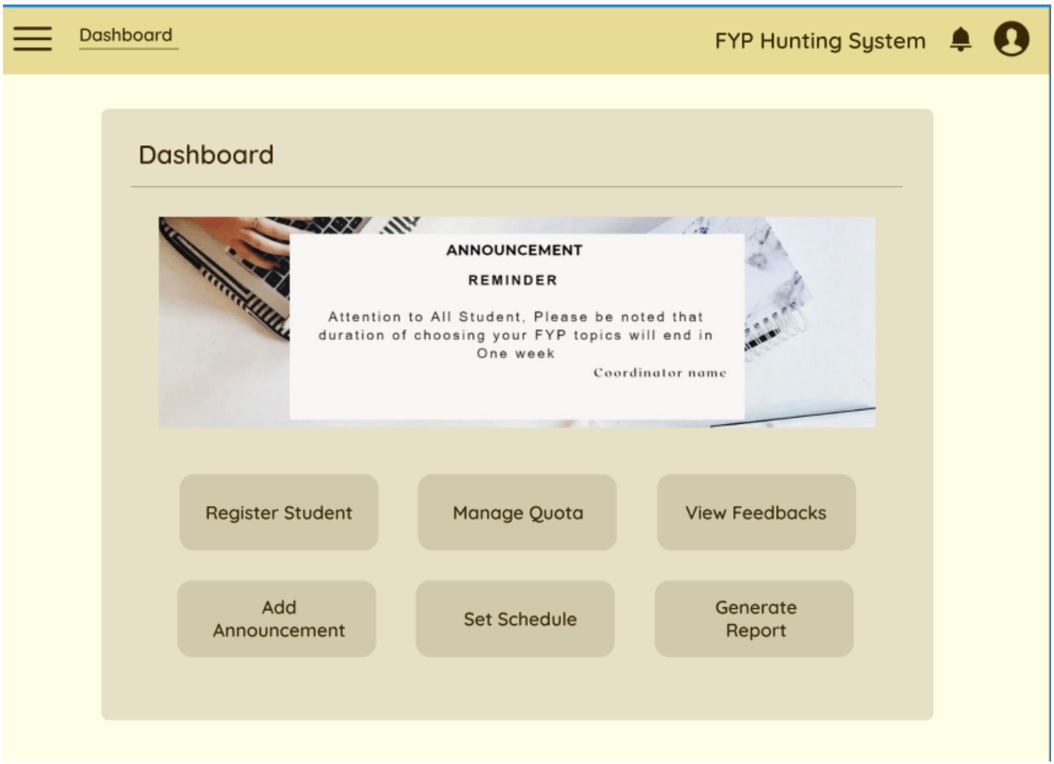
A-4.1: Sequence Diagram – Basic Flow

X

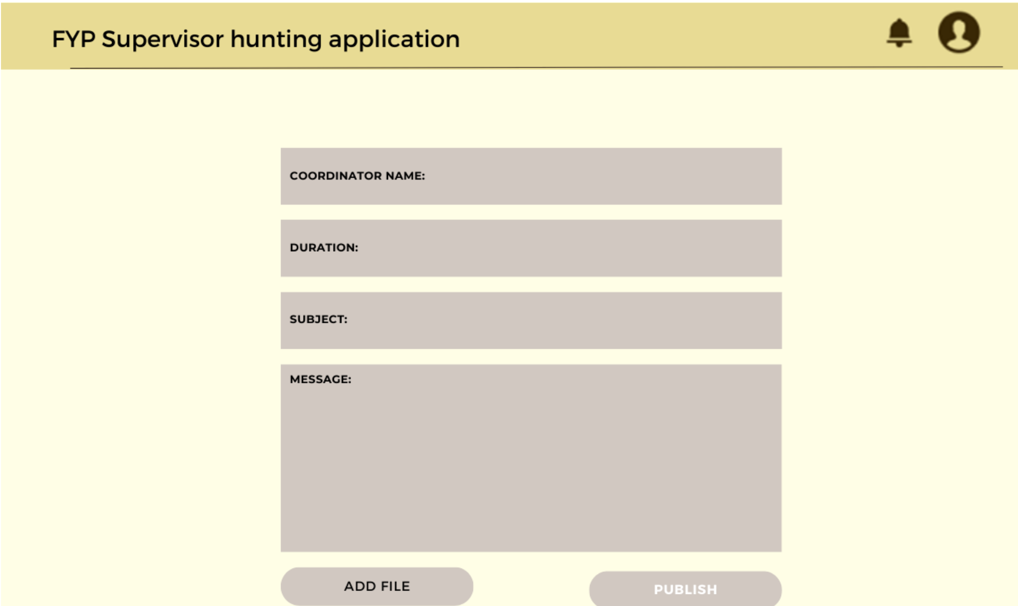
X

APPENDIX B
User Interfaces

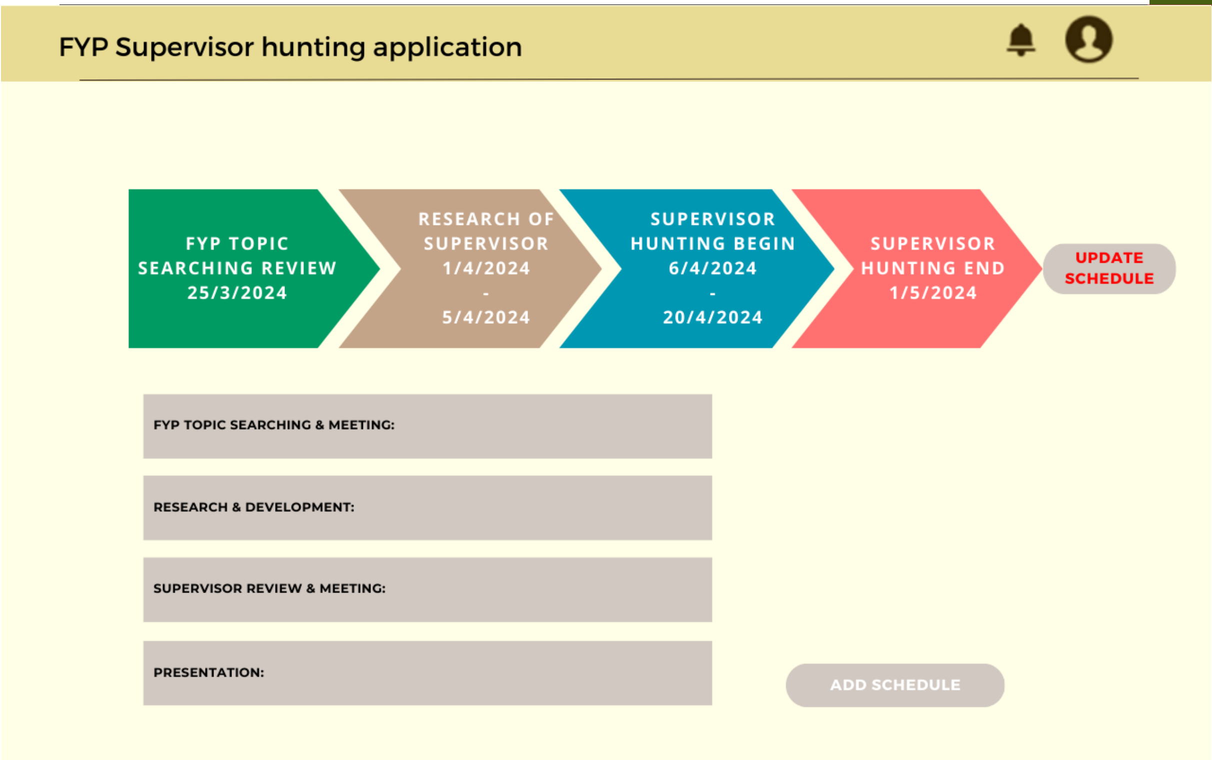
1. Manage Coordinator Activities (NUR ASYIKIN)



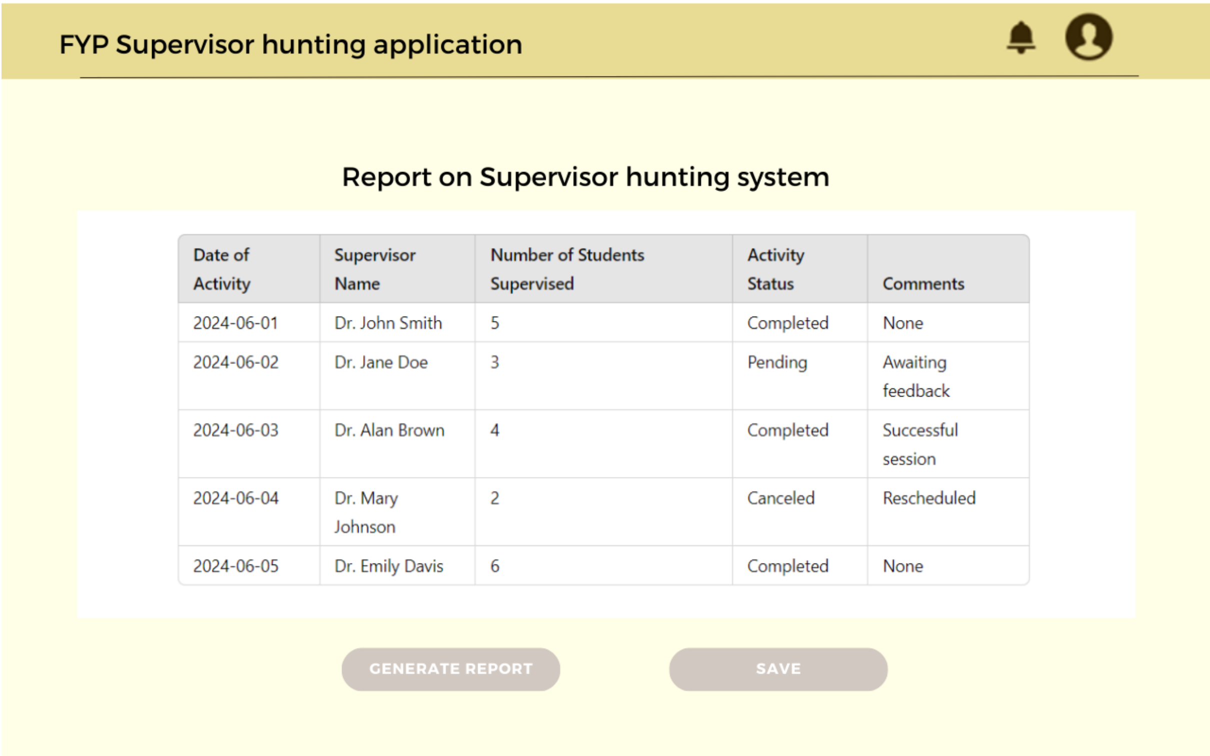
B-1.1: Coordinator Activities Dashboard



B-1.2: Add Announcement

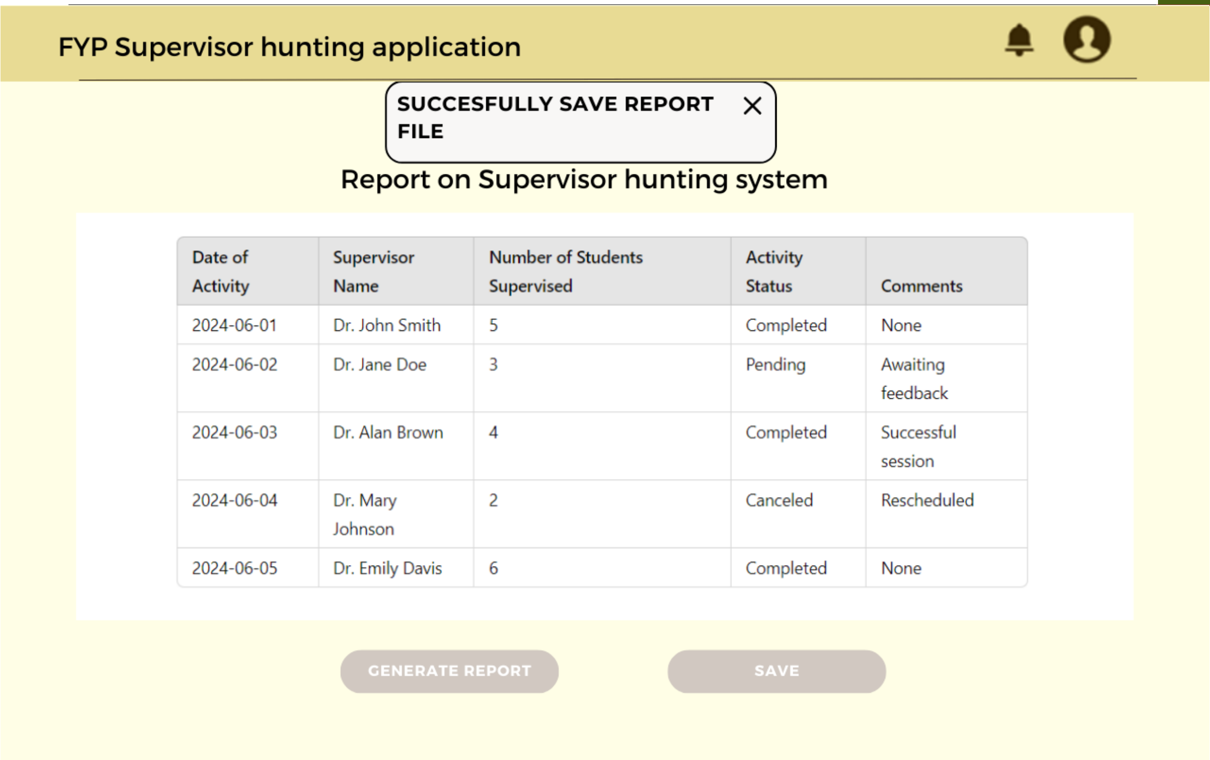


B-1.3: Schedule



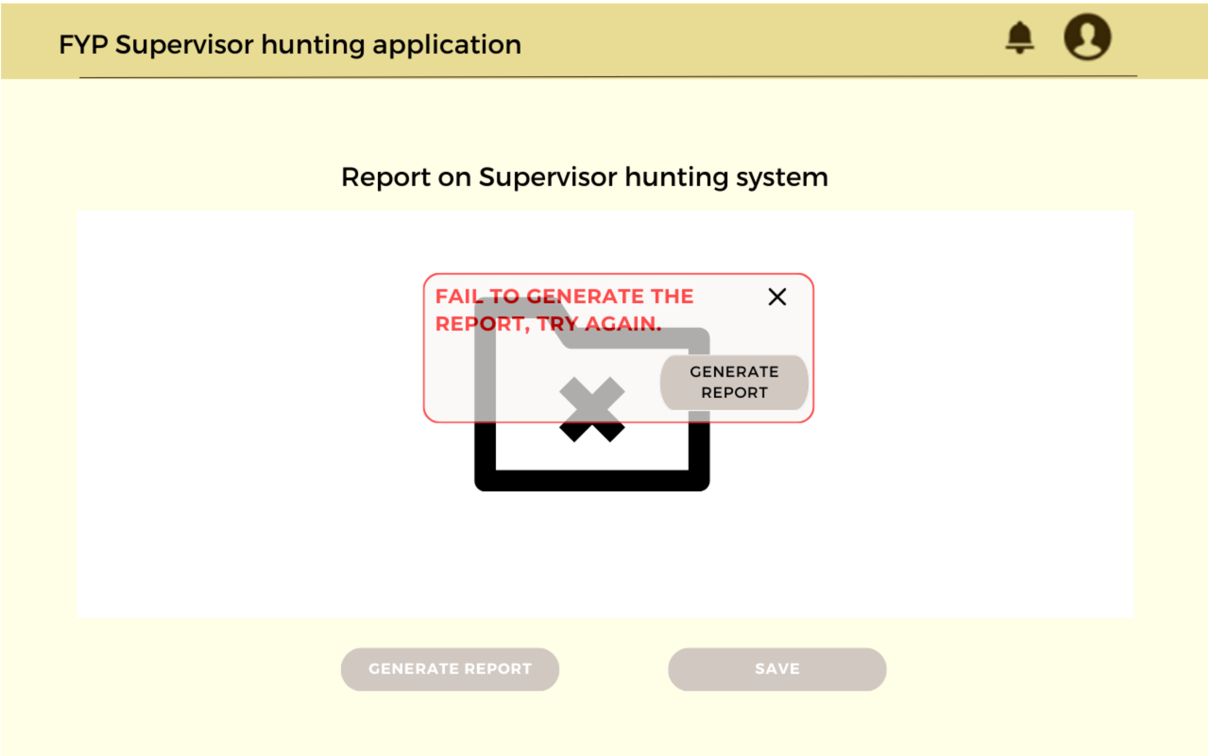
Date of Activity	Supervisor Name	Number of Students Supervised	Activity Status	Comments
2024-06-01	Dr. John Smith	5	Completed	None
2024-06-02	Dr. Jane Doe	3	Pending	Awaiting feedback
2024-06-03	Dr. Alan Brown	4	Completed	Successful session
2024-06-04	Dr. Mary Johnson	2	Canceled	Rescheduled
2024-06-05	Dr. Emily Davis	6	Completed	None

B-1.4: Generate Reports



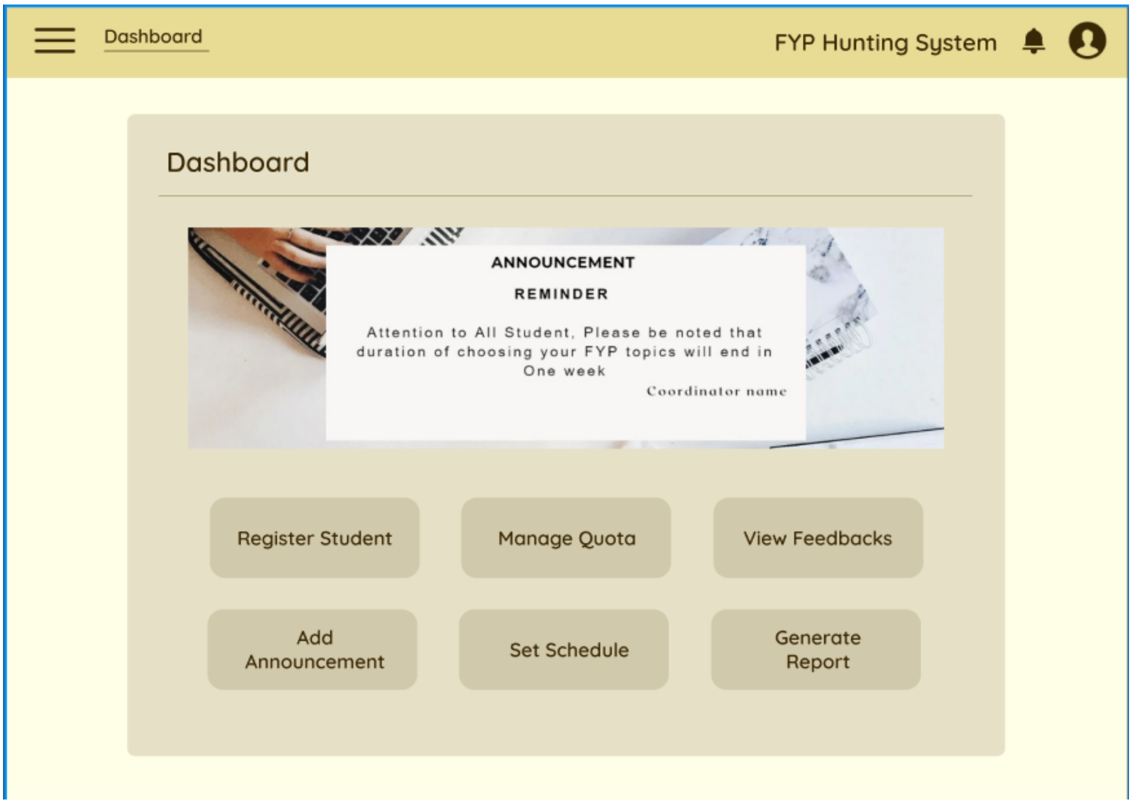
Date of Activity	Supervisor Name	Number of Students Supervised	Activity Status	Comments
2024-06-01	Dr. John Smith	5	Completed	None
2024-06-02	Dr. Jane Doe	3	Pending	Awaiting feedback
2024-06-03	Dr. Alan Brown	4	Completed	Successful session
2024-06-04	Dr. Mary Johnson	2	Canceled	Rescheduled
2024-06-05	Dr. Emily Davis	6	Completed	None

B-1.5: Save Report

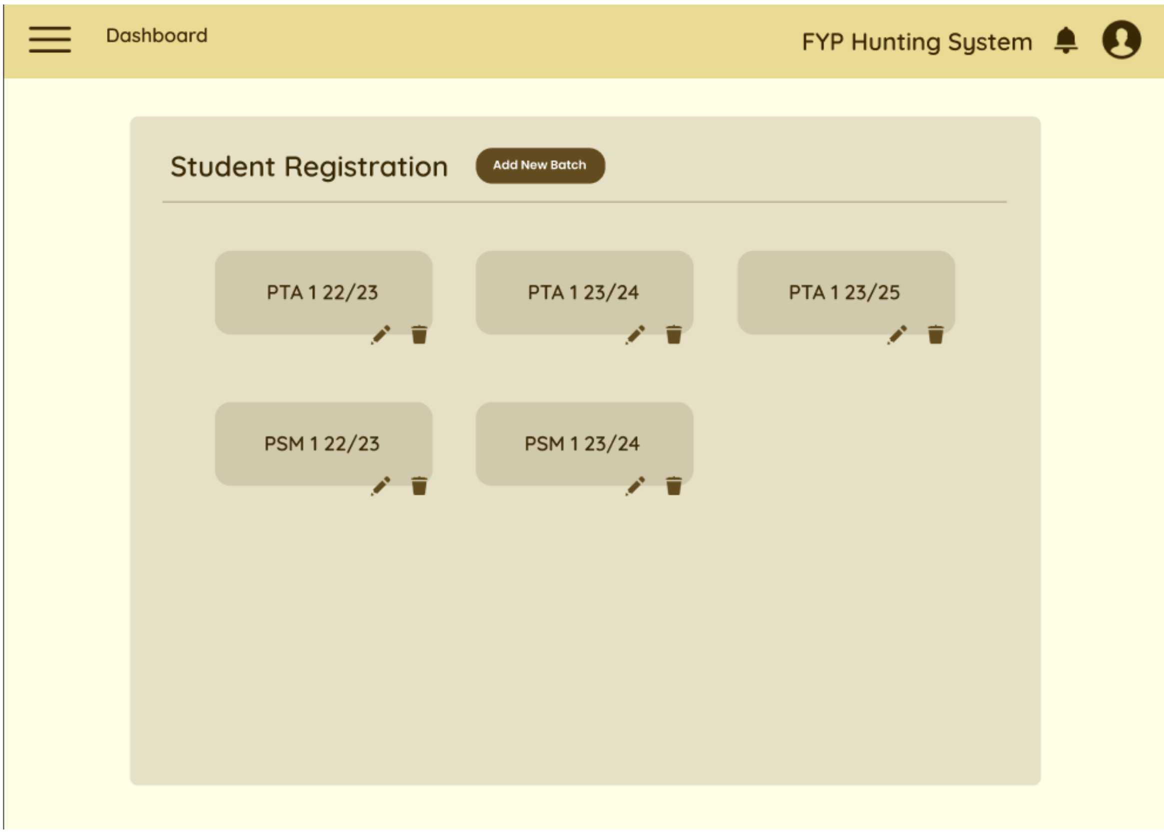


B-1.6: Generate reports error

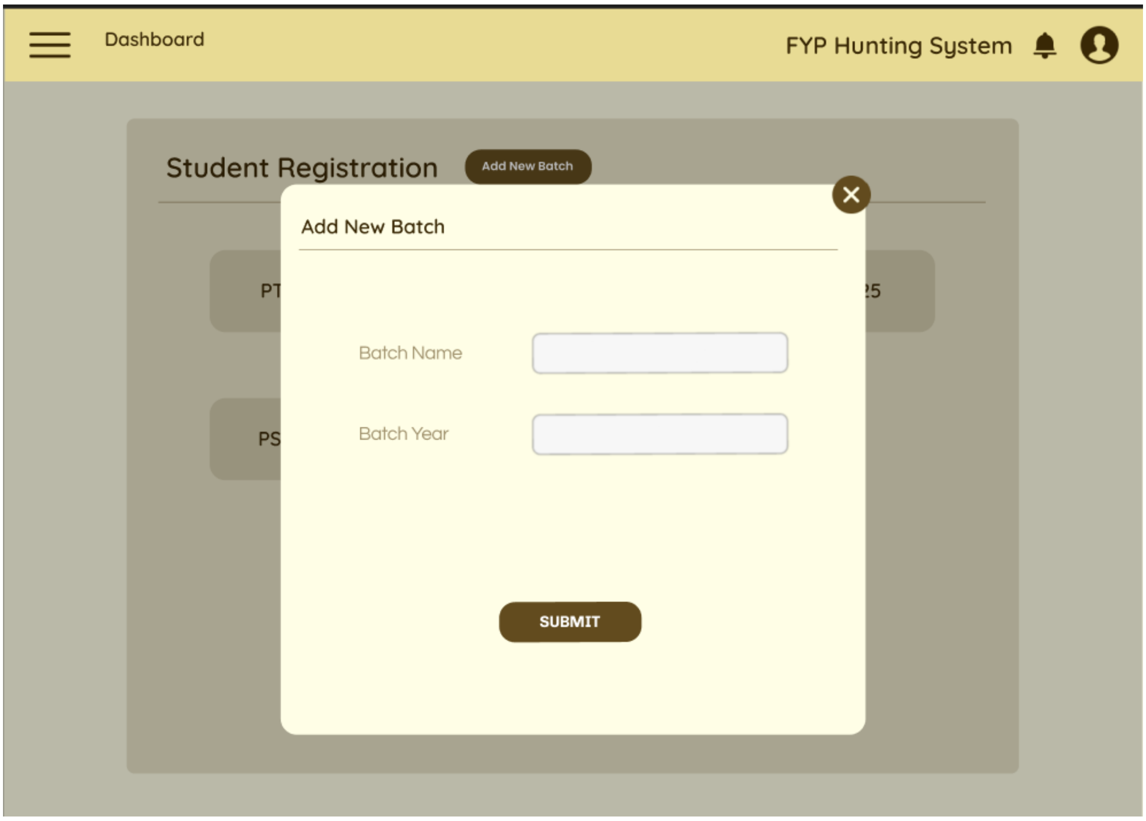
2. Manage Student (AMALA KARTHIGAYAN)



B-2.1: Coordinator Dashboard



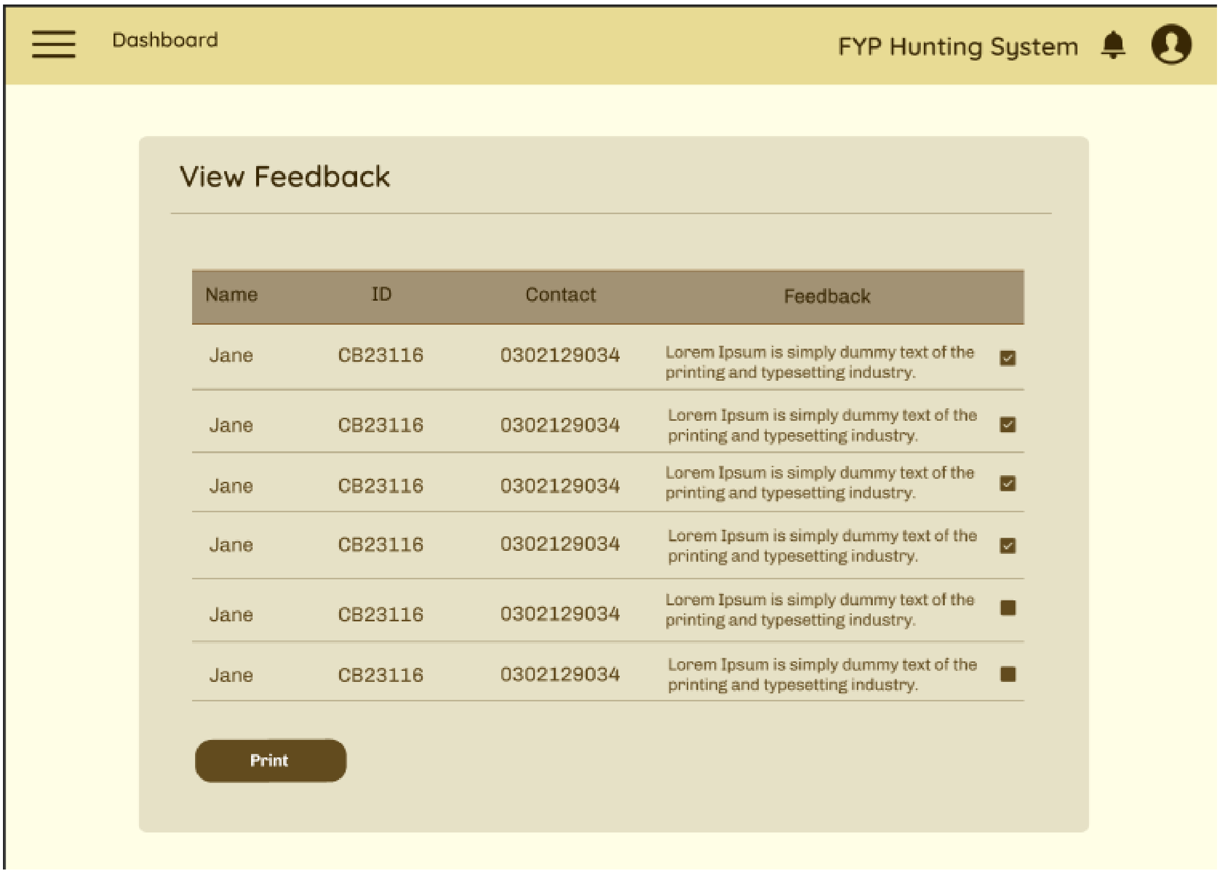
B-2.2: Student Registration Page



B-2.3: Add New Batch Page

B-2.4: Supervisee Quota

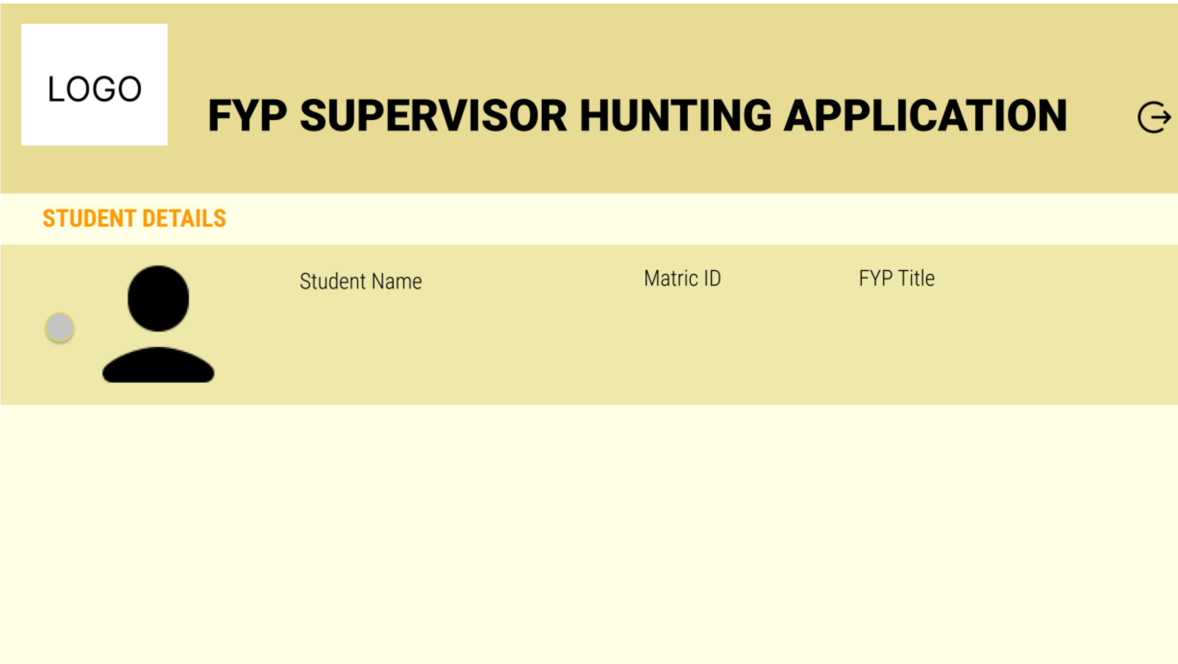
B-2.5: Student List Page



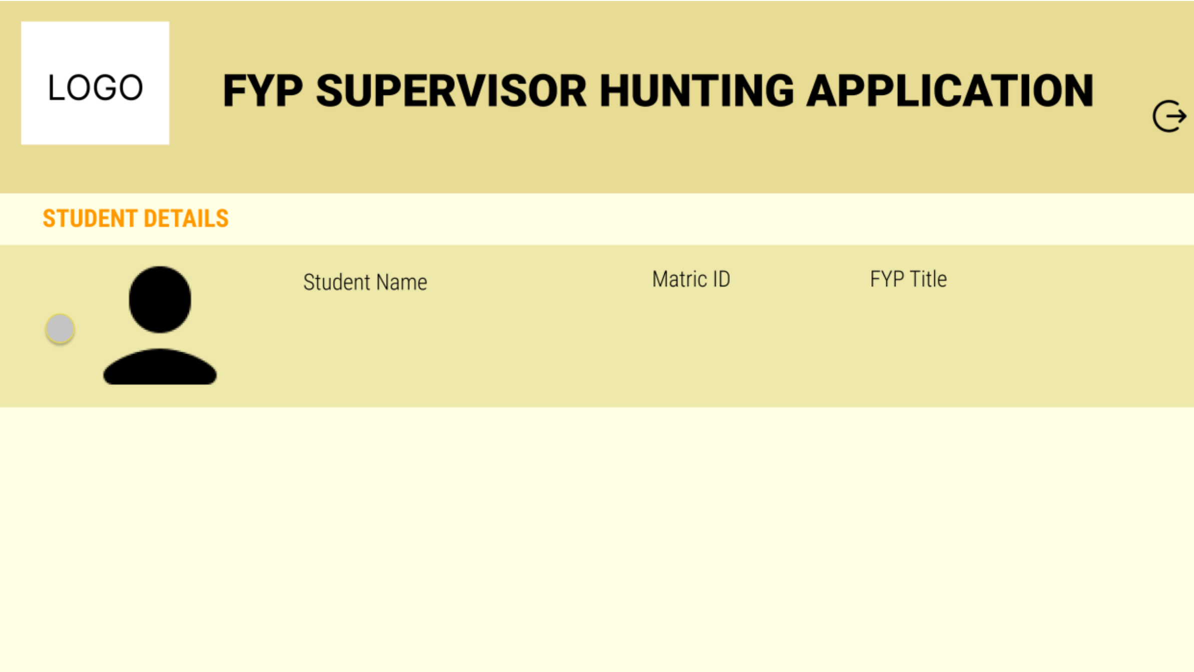
B-2.6: View Feedback

3. Manage Final Year Project & Manage Supervisor Hunting (MIRZA RUSYAIDI & NABILAH)

B-3.1: Lecturer Dashboard



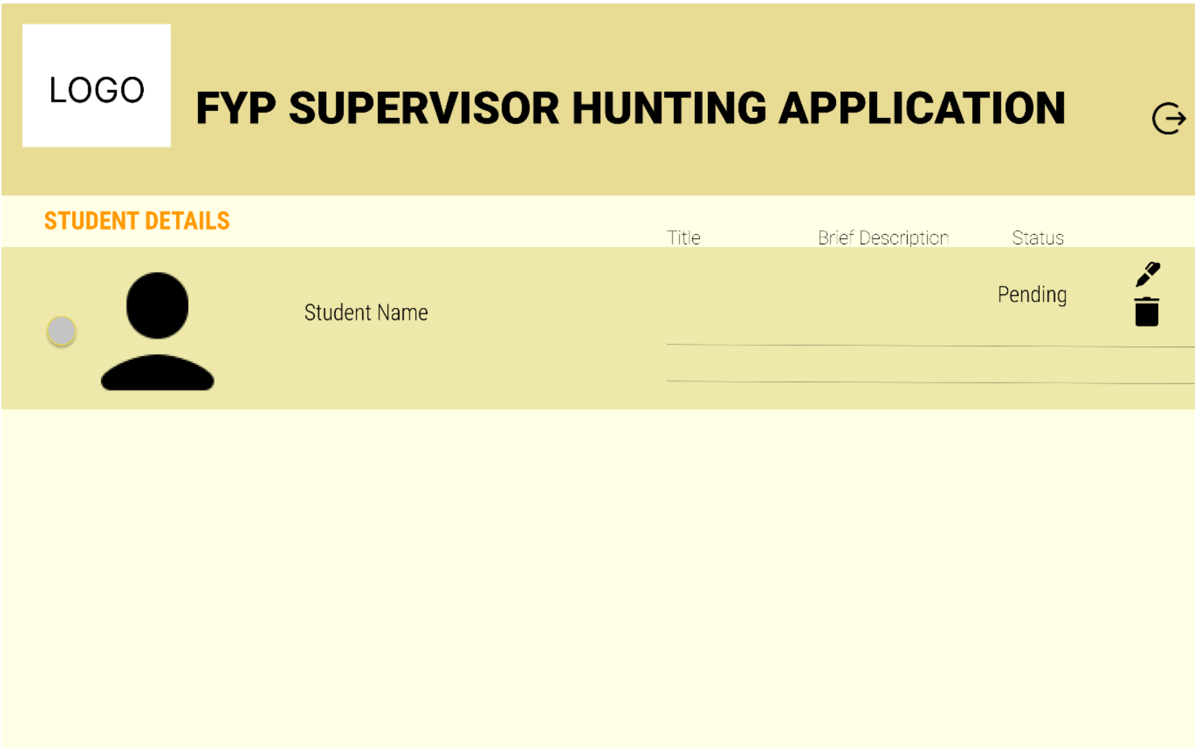
B-3.2: Supervisee Details



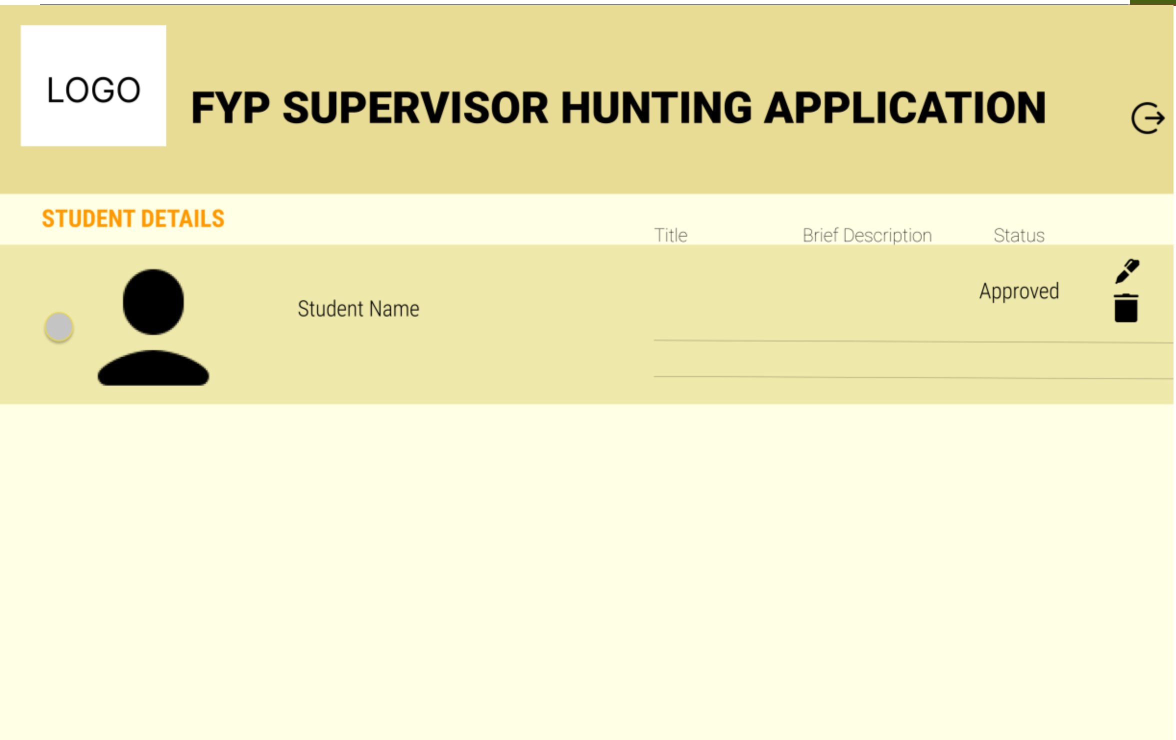
B-3.3: Final Year Project Interest Topic

B-3.4: FYP Interest Form

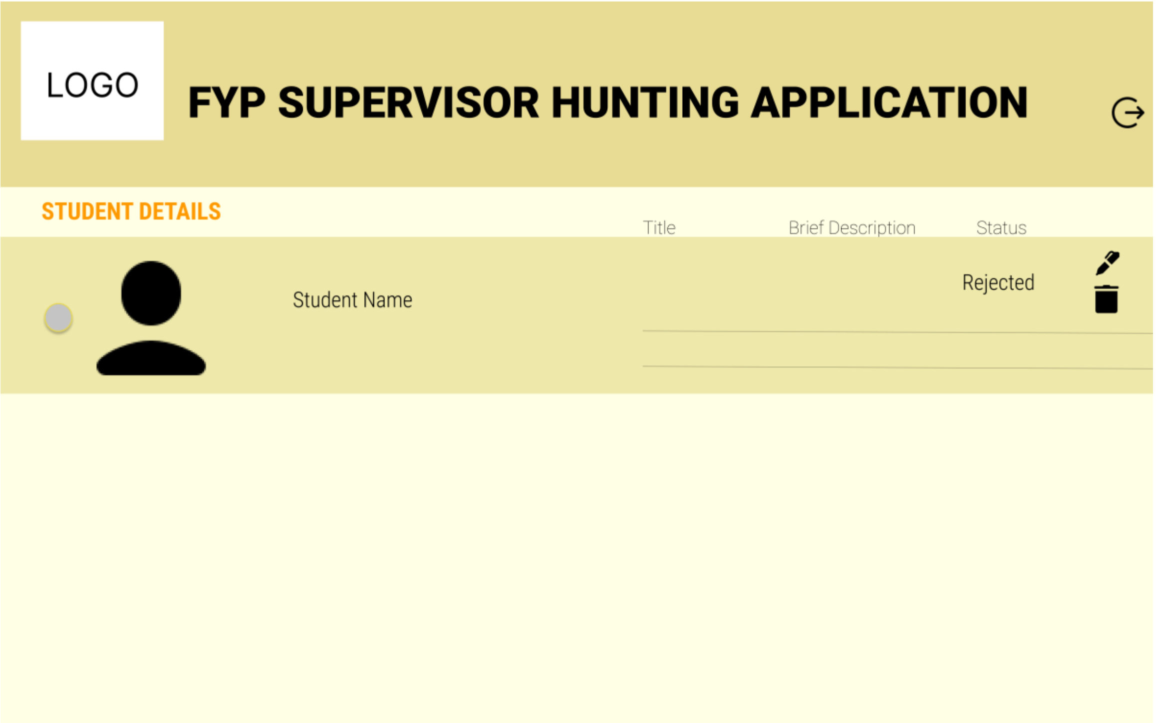
B-3.5: FYP Interest Form Update



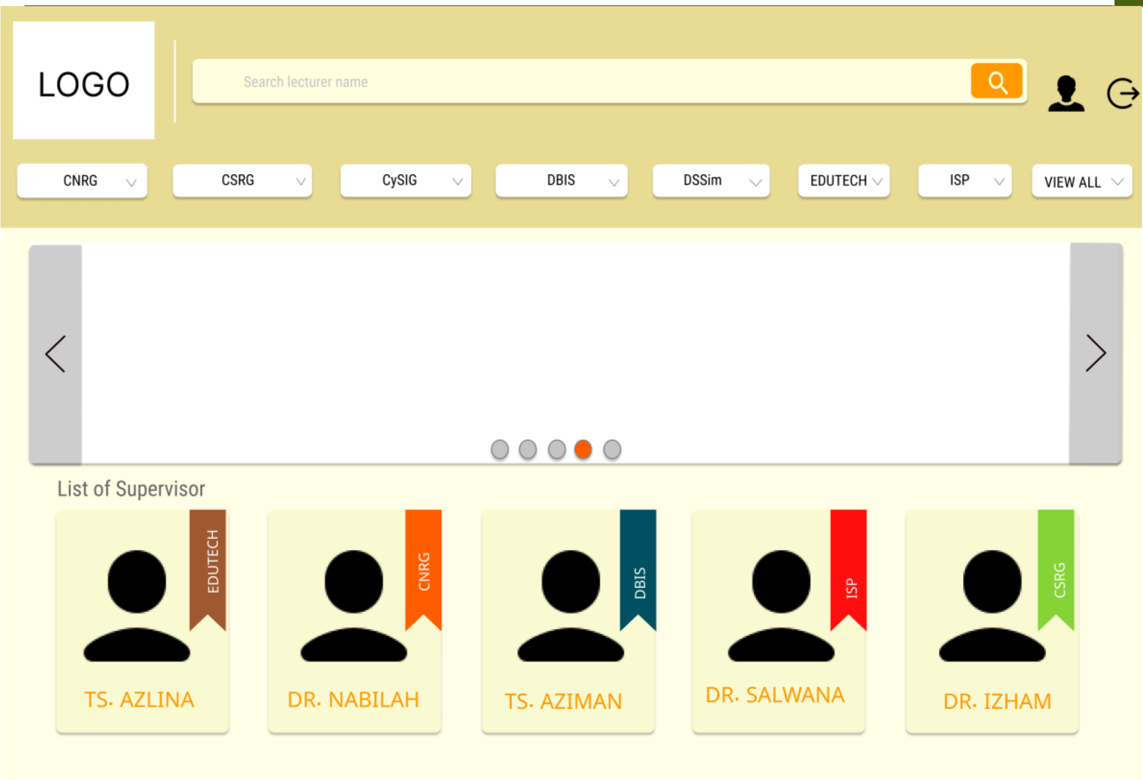
B-3.6: Supervisee Request Approval Form (Pending)



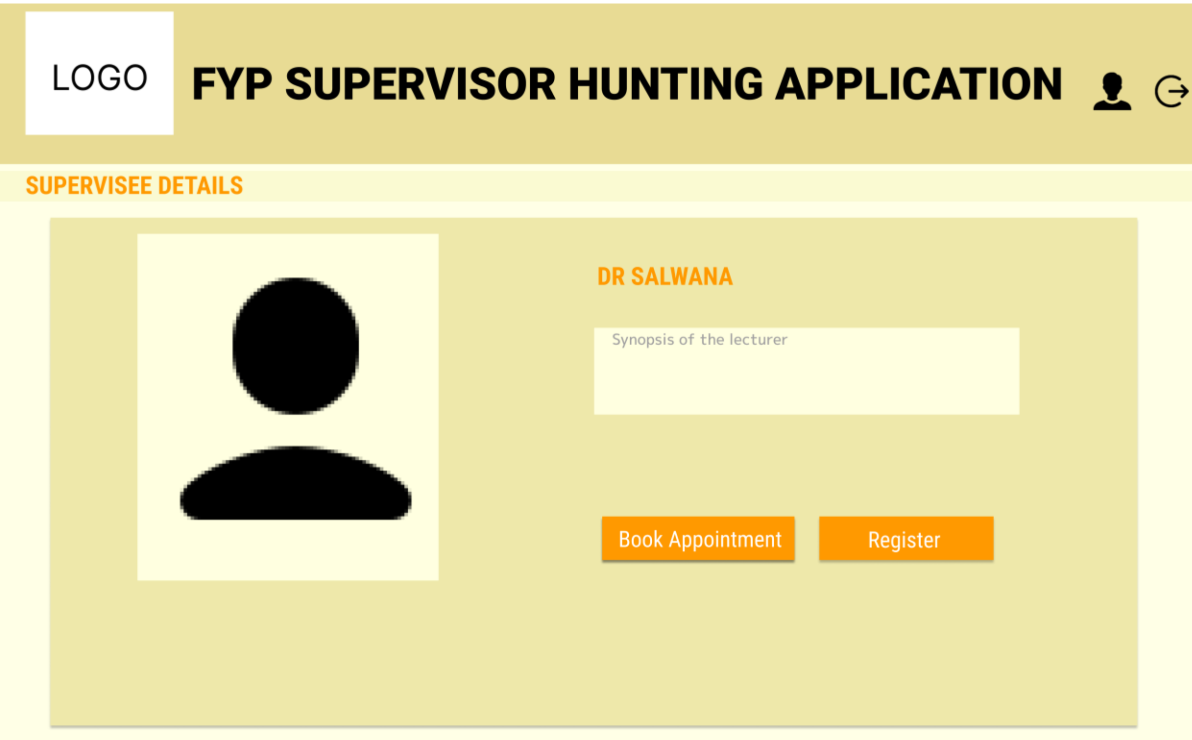
B-3.7: Supervisee Request Approval Form (Approved)



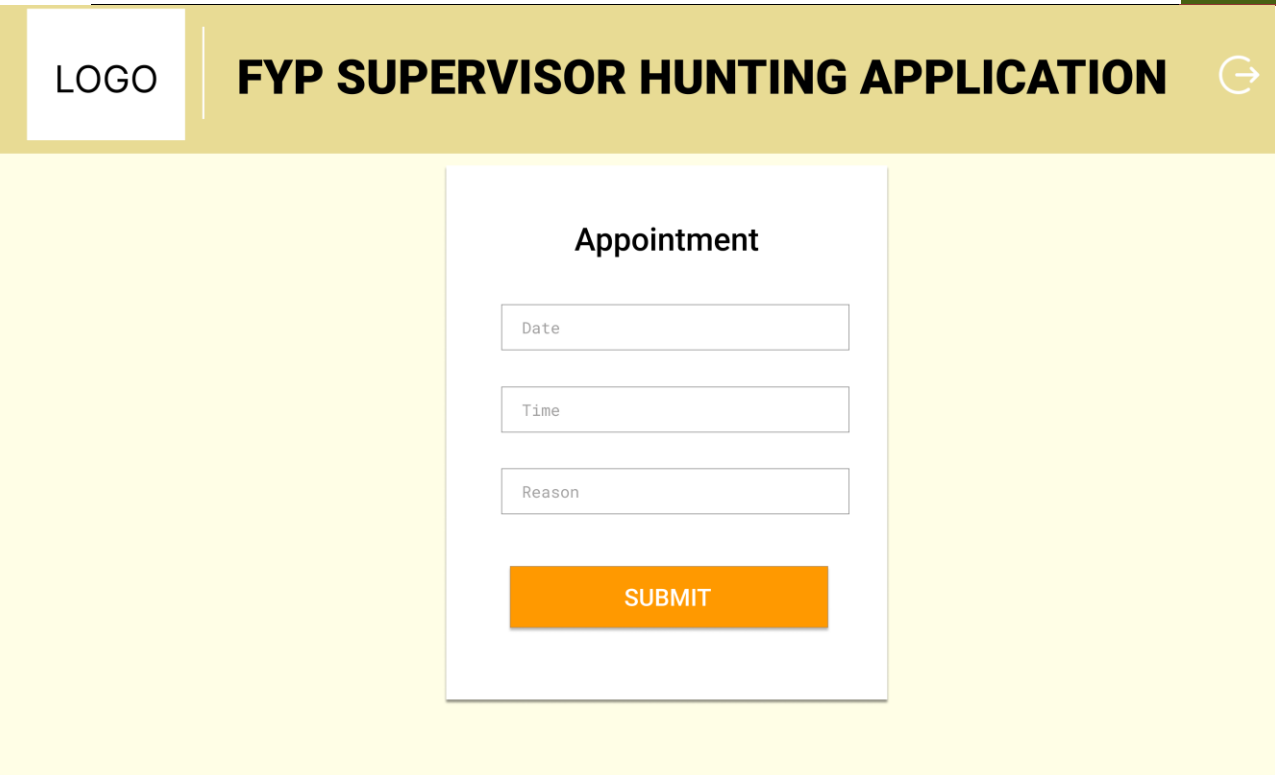
B-3.8: Supervisee Request Approval Form (Rejected)



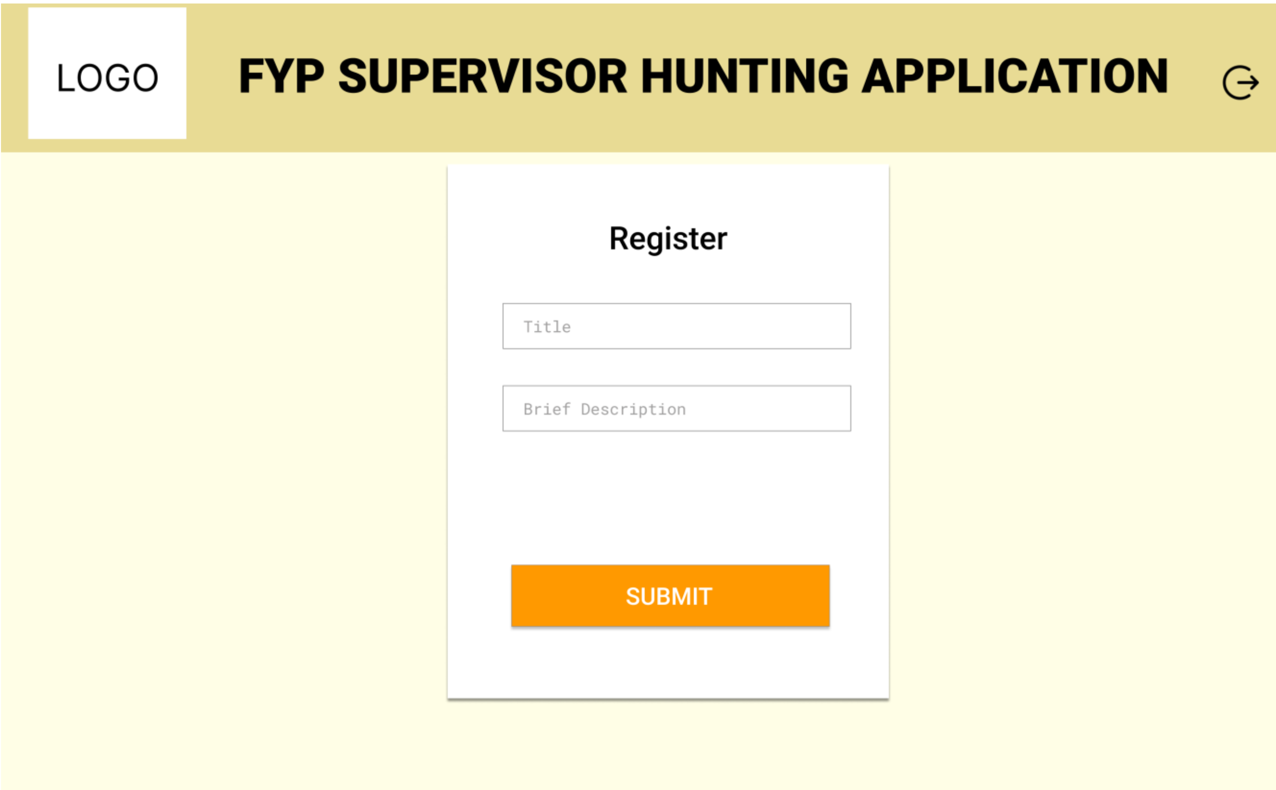
B-3.9: Student Dashboard



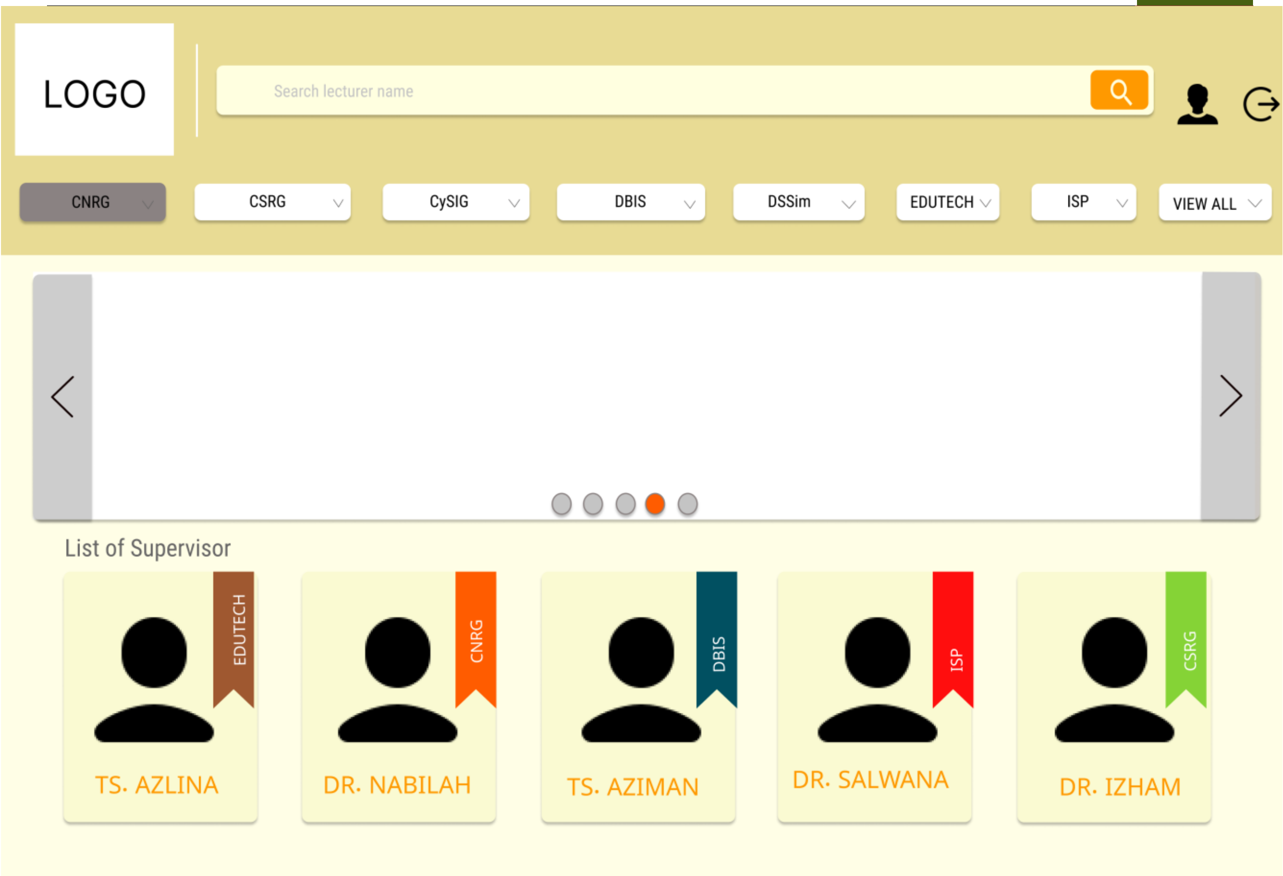
B-3.10: Supervisor Details



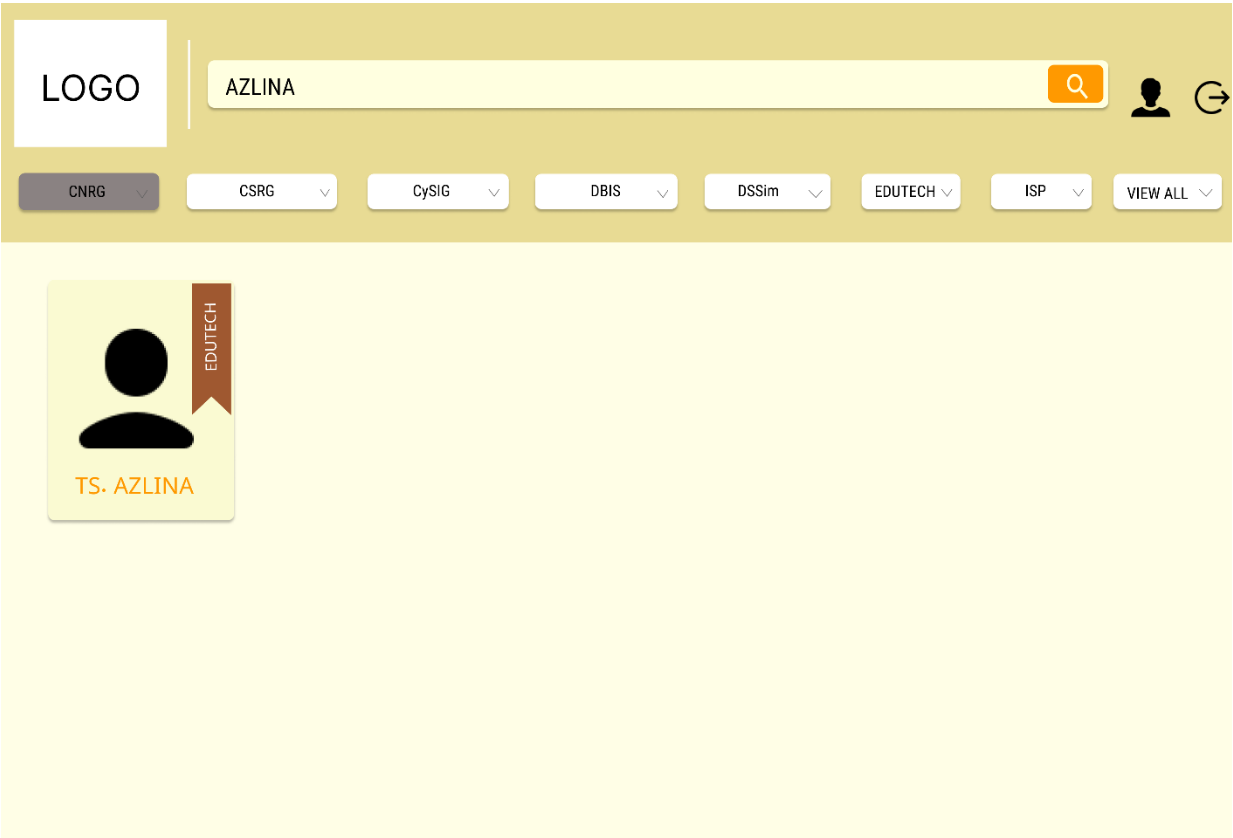
B-3.11: Book Appointment



B-3.12: Supervisor Registration

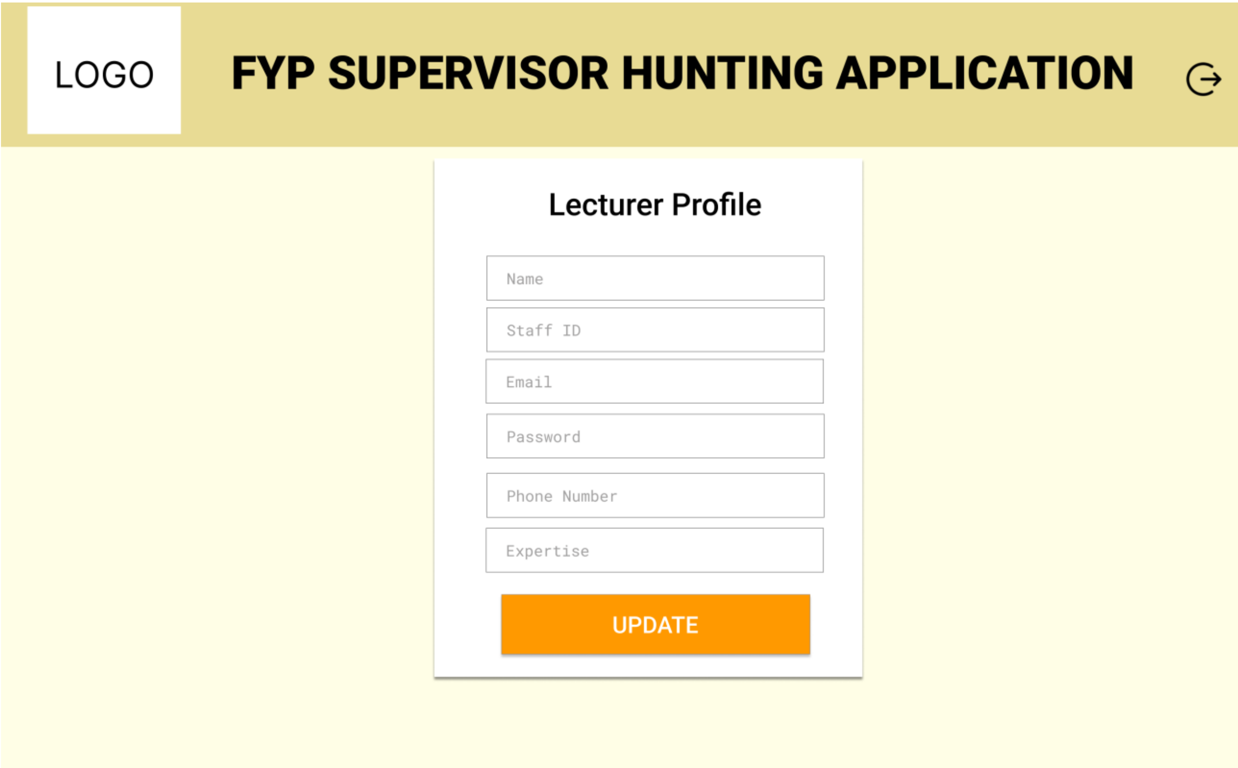


B-3.13: Sort Research Group



B-3.14: Search Lecturer Name

B-3.15: Student Profile



B-3.16: Lecturer Profile